

MicroMerit Portal: A Blockchain and AI-Powered Credential Aggregator Platform

A PROJECT REPORT

Submitted by

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Abstract

The rapid proliferation of digital learning platforms has created a critical need for a unified, tamper-proof system to manage and verify academic and vocational credentials. Traditional credential management relies on paper certificates or centralized databases, both of which are susceptible to forgery, data loss, and lack of interoperability. This project, titled MicroMerit Portal, addresses these limitations by designing and implementing a Credential Aggregator Platform powered by Blockchain, IPFS, and Artificial Intelligence.

MicroMerit Portal is a multi-role web application supporting four user types: Learners (students), Issuers (institutions), Employers (organizations), and Administrators. The platform enables institutions to issue verifiable digital credentials — either individually through a web portal, in bulk via ZIP upload, or programmatically through a REST API. Each issued credential is cryptographically anchored on the Ethereum Sepolia testnet using a Solidity smart contract (CredentialRegistry), with the credential PDF stored on IPFS for decentralized, content-addressed access.

The platform implements two complementary verification methods. The primary method uses blockchain-backed cryptographic proof: a canonical JSON representation of the credential is built with all keys sorted alphabetically, hashed using SHA-256, and the resulting data_hash is stored on-chain. Verifiers can confirm a credential's authenticity by re-computing this hash and querying the smart contract. The secondary method uses Google Gemini 2.5 Flash for AI-powered document comparison — useful when only a scanned or photographed copy is available. Additionally, the platform features an AI-powered OCR pipeline (Tesseract + Groq Llama 3.1 8B) for extracting credential metadata from uploaded documents, an external credential sync system using a provider-agnostic connector pattern (supporting Udemy, Credly-like, and other external providers), and an AI chatbot for employers to query learner credentials using natural language.

The system is built as six independent microservices: a Node.js/Express/TypeScript main backend, a Python/FastAPI AI service, a TypeScript/Hardhat blockchain service, a React frontend, a React-based admin dashboard, and a credential provider abstraction layer.

PostgreSQL serves as the relational database and Redux Toolkit manages frontend state. The platform demonstrates the practical viability of integrating blockchain immutability, decentralized storage, and AI intelligence into a cohesive credential management ecosystem.

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List of Symbols, Abbreviations and Nomenclature

Abbreviation	Full Form
IPFS	InterPlanetary File System
API	Application Programming Interface
JWT	JSON Web Token
OTP	One-Time Password
SHA-256	Secure Hash Algorithm 256-bit
UUID	Universally Unique Identifier
DFD	Data Flow Diagram
ER	Entity Relationship
OCR	Optical Character Recognition
LLM	Large Language Model
REST	Representational State Transfer
ORM	Object Relational Mapper
UI	User Interface
QR	Quick Response
CRUD	Create, Read, Update, Delete

CHAPTER 1: INTRODUCTION

1.1 Overview

The emergence of digital credentials and online learning platforms has transformed how individuals acquire and demonstrate skills. In today's fast-paced knowledge economy, learners accumulate certificates from diverse sources — universities, MOOCs, vocational institutes, and online learning platforms. However, these credentials are typically stored as paper certificates or PDFs on centralized servers, creating risks of forgery, loss, and inability to verify authenticity efficiently. Employers and institutions lack a single trusted interface to verify credentials from multiple providers.

MicroMerit Portal is a Credential Aggregator Platform that addresses these challenges by combining Blockchain technology, IPFS decentralized storage, and Artificial Intelligence into a unified credential management ecosystem. The platform supports four user roles — Learner, Issuer, Employer, and Admin — and is implemented as a microservices architecture comprising six independent services.

1.2 Motivation

Paper-based credentials and unmanaged digital PDFs continue to dominate many educational and vocational ecosystems. Such credentials can be lost, damaged, or duplicated, and PDFs can be altered using common editing tools without leaving obvious traces. Additionally, learners typically store certificates across emails, cloud drives, or platform-specific dashboards, resulting in no practical cross-platform aggregation of a complete skill portfolio.

From an employer's perspective, verification often involves time-consuming manual processes such as contacting institutions or requesting re-validation emails, which slows hiring and increases administrative overhead. While platforms such as LinkedIn and Credly provide digital representations of achievements, they are largely siloed and do not integrate well with credentials from other providers or institutions, limiting their usefulness for comprehensive candidate evaluation.

This project is motivated by the opportunity to combine three complementary technologies: blockchain can provide immutable, public proof of credential integrity without requiring trust in a central authority; IPFS can provide permanent, content-addressed storage for credential documents; and AI can support flexible verification even when an exact original file is unavailable (e.g., scans or photographs). Personally, the motivation was to build a practically useful system for the Indian skill-development ecosystem that reduces verification friction and helps learners present a credible, unified portfolio to employers.

1.3 Problem Statement

The existing credential management ecosystem suffers from the following critical limitations:

1. Credentials issued by different institutions are stored in isolated systems with no standardized aggregation layer for learners.
2. Verification of credentials relies on manual processes (contacting institutions), which are time-consuming and prone to error.
3. Digital certificates in PDF format can be trivially forged by editing metadata, with no cryptographic proof of authenticity.
4. Learners have no single trusted public profile to present their complete credential portfolio to employers.
5. Employers lack AI-assisted tools to intelligently query and compare candidate skill profiles.

1.4 Proposed Solution

MicroMerit Portal addresses each of the above limitations through the following key capabilities:

Table 1.1: Problem-Solution Mapping

Problem	Solution	Technology Used
Siloed credentials	External Sync via connector pattern	Udemy/Credly-like provider adapters
Manual verification	Automated dual-method verification	Blockchain smart contract + Google Gemini AI
PDF forgery	PDF checksum + canonical hash on blockchain	SHA-256, Ethereum, pdf-lib
No unified profile	Public learner profile pages (/p/:slug)	React, PostgreSQL, QR code

No AI for employers	AI chatbot + candidate comparison	Groq Llama 3.1 8B, Google Gemini
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1.5 Existing Systems and Limitations

Several platforms address parts of digital credentialing. Credly focuses on digital badges and is commonly used in US-centric professional certification ecosystems, but it does not provide blockchain-based proof of issuance. LinkedIn Learning attaches certifications to a user profile, but it does not provide cryptographic verification of certificate integrity. In addition, many institutions provide certificate portals that issue static PDFs, which can be shared but do not include robust tamper detection or standardized verification workflows.

Despite these platforms, none provides: (a) blockchain-anchored immutability, (b) AI-powered flexible verification, (c) bulk credential issuance with adapter-based file processing, and (d) an employer-facing AI chatbot for credential queries. MicroMerit Portal combines these requirements into a single platform, enabling multi-provider aggregation, issuance, and verification workflows.

Table 1.2: Comparison of Existing Systems

Platform	Blockchain Proof	AI Verification	Multi-Provider Sync	Employer Chatbot	Bulk Issuance
Credly	No	No	No	No	No
LinkedIn Learning	No	No	Yes	No	No
Institutional Portals	No	No	No	No	No
MicroMerit Portal	Yes	Yes	Yes	Yes	Yes

1.6 Scope of the Project

The scope of MicroMerit Portal includes the following functional modules and deliverables:

- Multi-role web application (Learner, Issuer, Employer, Admin)
- Single and bulk credential issuance with blockchain anchoring and IPFS storage
- Two-method credential verification: blockchain cryptographic proof and AI document comparison
- External credential sync from providers (Udemy, Credly-like, and others) via connector pattern
- Public learner profiles and shareable credential pages with QR codes
- Employer tools: candidate search, AI chatbot, candidate comparison, bulk ZIP verification
- Admin portal for issuer approval and credential monitoring
- **Future Scope:** Skill Pathway AI (roadmap generation based on current credentials), AI-generated skill profiles, course recommendations

1.7 Summary

This chapter introduced the concept of micro-credentials and the challenges of fragmented, forgeable, and hard-to-verify credential records. It presented the motivation for a unified system that supports trusted verification and aggregation across multiple providers. The chapter then defined the problem statement and mapped each limitation to the proposed solution capabilities of MicroMerit Portal. Finally, it described the project scope. The subsequent chapters describe the literature review and technologies (Chapter 2), system modeling and design (Chapter 3), implementation and results (Chapter 4), and conclusions with future scope (Chapter 5).

CHAPTER 2: RELATED WORK AND BACKGROUND

2.1 Introduction

This chapter presents related work and background concepts relevant to MicroMerit Portal. It includes a brief literature review on blockchain-based credentialing and decentralized storage, followed by requirement analysis covering both functional and non-functional requirements. Finally, it describes the key technologies used to implement the system and summarizes the overall system architecture.

2.2 Existing Systems and Literature Review

Blockchain-based credential systems have been studied extensively. MIT Media Lab's Blockcerts project demonstrated that blockchain can be used to issue and verify academic certificates with cryptographic proof, establishing the foundational principle that a hash of a

credential stored on a public blockchain provides immutable verification without requiring the issuing institution to remain online [1].

IPFS has been explored as a decentralized storage layer for documents where integrity is critical. Prior work highlights that IPFS provides content-addressed, decentralized storage where a file's content hash (CID) serves as both its address and its integrity proof. This is particularly suitable for credential PDFs because the CID changes if even a single byte of the document is modified [2].

AI-based credential extraction and verification has gained relevance due to the widespread use of scanned documents and heterogeneous certificate formats. OCR-based metadata extraction from credential documents has been explored in document intelligence systems, and tools such as Tesseract OCR combined with large language models can extract structured data (e.g., name, institution, date, marks) from unstructured PDFs and images with high accuracy [3].

Multi-modal AI, particularly vision-language models such as Gemini, enables document comparison beyond byte-level matching. Unlike hashing, vision-language models can perform semantic comparison of documents, tolerating scan artifacts, image quality variation, and formatting differences while still detecting content-level mismatches such as altered names or dates. This makes AI verification a practical complement to cryptographic verification in real-world workflows [4].

Enterprise integration literature also motivates the connector/adaptor approach used for credential aggregation. The connector pattern allows a system to communicate with multiple external data sources through a common interface, abstracting provider-specific authentication, request formats, and data models behind a uniform API. This approach is used extensively in enterprise aggregation systems where multiple providers must be supported without rewriting core business logic [5].

2.3 Requirement Analysis

2.3.1 Functional Requirements

The primary functional requirements of the system are summarized in Table 2.1.

Table 2.1: Functional Requirements

Req. ID	User Role	Requirement
FR-01	Learner	Register and log in via Email+OTP or Google OAuth
FR-02	Learner	View all issued credentials in a personal dashboard
FR-03	Learner	Sync credentials from external providers (Udemy, Credly-like)
FR-04	Learner	Share a public profile page and individual credential pages
FR-05	Issuer	Register (pending admin approval) and log in
FR-06	Issuer	Issue a single credential (PDF + metadata) to a learner
FR-07	Issuer	Bulk issue credentials via ZIP file upload
FR-08	Issuer	Manage REST API keys for programmatic credential issuance
FR-09	Employer	Search learners by skill, certificate title, issuer name, location
FR-10	Employer	Verify a credential by ID, tx_hash, IPFS CID, PDF upload, or AI comparison
FR-11	Employer	Use AI chatbot to ask natural-language questions about a learner's credentials
FR-12	Employer	Compare up to 3 candidates side-by-side using AI, export results to CSV
FR-13	Admin	Approve or reject issuer registration requests
FR-14	Admin	Monitor all credentials across the platform
FR-15	Admin	Control external credential sync settings per provider
FR-16	System	Anchor every issued credential's data_hash on Ethereum Sepolia blockchain
FR-17	System	Store credential PDFs on IPFS with content-addressed CIDs

2.3.2 Non-Functional Requirements

Table 2.2: Non-Functional Requirements

Category	Requirement
Security	JWT-based stateless authentication; bcrypt password hashing; OTP verification for sensitive actions
Integrity	Blockchain-anchored SHA-256 data_hash ensures credential immutability
Availability	Microservices architecture allows independent deployment and restart of services
Scalability	Microservices architecture allows independent deployment; asynchronous background workers decouple credential anchoring from HTTP response cycle
Usability	Role-based dashboards with responsive React UI; QR code sharing for credentials
Interoperability	Connector pattern abstracts external credential provider APIs behind a uniform interface
Auditability	All employer credential access actions logged in employer_activity_log table

2.4 Technologies Used

MicroMerit Portal is implemented using a modern full-stack architecture with blockchain and AI extensions. This section briefly describes the primary frameworks, platforms, and tools used to build the backend services, frontend applications, and verification pipeline.

2.4.1 Node.js and Express.js

Node.js is a server-side JavaScript runtime designed for scalable network applications with non-blocking I/O [6]. In MicroMerit Portal, Node.js is used to implement the main backend server responsible for authentication, credential issuance, verification orchestration, and role-based APIs. Express.js provides RESTful routing and middleware support to structure these APIs in a maintainable manner [7].

2.4.2 TypeScript

TypeScript extends JavaScript with static type checking, reducing runtime errors and improving long-term code maintainability [8]. The main backend and the blockchain service are written in TypeScript to ensure consistent data models and safer refactoring as the platform evolves.

2.4.3 React.js and Vite

React.js is a component-based UI library that uses a virtual DOM for efficient rendering of interactive web interfaces [9]. Vite is used as the frontend build tool to provide fast development startup and hot module replacement. Redux Toolkit with Redux Persist manages global state and persists authentication tokens across page refreshes.

2.4.4 PostgreSQL and Prisma ORM

PostgreSQL is an open-source relational database that provides ACID compliance and support for advanced indexing features such as JSONB indexing [10]. Prisma ORM provides a type-safe database client generated from the schema definition, simplifying database access from TypeScript services. JSONB columns store credential metadata (tags, awarding bodies, canonical JSON) with GIN indexing for efficient querying.

2.4.5 Background Job Processing

Blockchain anchoring and IPFS upload are handled as background tasks using Node.js's built-in `setImmediate`, dispatched immediately after the credential is saved to the database. This ensures the HTTP response is returned to the issuer without waiting for on-chain confirmation, keeping the user-facing API responsive.

2.4.6 Python, FastAPI, and Tesseract OCR

The AI service is implemented in Python using the FastAPI framework for high-performance asynchronous HTTP endpoints [11]. Tesseract OCR extracts raw text from uploaded credential PDFs and images [12]. The extracted text is then sent to the Groq API running Llama 3.1 8B to parse structured metadata fields such as learner name, institution, course title, and issuance date.

2.4.7 Groq API (Llama 3.1 8B)

Groq provides ultra-low-latency inference for open-source large language models through a hosted API [13]. In this project, Llama 3.1 8B is used both for OCR text post-processing and as the backend LLM for the employer AI chatbot.

2.4.8 Google Gemini 2.5 Flash

Google Gemini 2.5 Flash is a multimodal vision-language model capable of understanding and comparing document images [14]. In MicroMerit Portal, it is used for AI-powered credential verification: the original credential PDF (fetched from IPFS) and the user-submitted document copy are both provided to Gemini, which performs a field-by-field semantic comparison. The model returns a match verdict, confidence score (0–100%), list of mismatched fields, and a one-sentence summary.

2.4.9 Solidity and Hardhat (Ethereum)

Solidity is a statically-typed programming language for writing Ethereum smart contracts [15]. The CredentialRegistry smart contract is deployed on the Ethereum Sepolia testnet and stores a mapping from `credential_id` to a struct containing the SHA-256 `data_hash`, IPFS CID, issuer wallet address, and timestamp. Hardhat is used for compiling, testing, and deploying the contract, while Ethers.js is used from Node.js to interact with the deployed contract [16].

2.4.10 IPFS

IPFS (InterPlanetary File System) is a peer-to-peer distributed file system where content is addressed by its cryptographic hash (CID) rather than by location [17]. Credential PDFs uploaded to IPFS receive a unique CID; any modification to the file changes the CID, providing built-in integrity verification.

2.4.11 Tailwind CSS

Tailwind CSS is a utility-first CSS framework used to style both the main frontend and the admin dashboard with a consistent, responsive design [18].

2.5 System Architecture Overview

The MicroMerit Portal follows a microservices architecture comprising six independent services communicating over HTTP. The Main Backend API (Node.js, Port 3000) acts as the central orchestrator, routing authenticated requests to the AI Service, Blockchain Service, and Credential Provider Adapter. The two frontend applications (Main App on Port 5173 and Admin Dashboard on Port 5174) communicate with the backend exclusively through REST API calls.

Data persistence is split across three layers: PostgreSQL stores all relational data (users, credentials, batches), IPFS stores the credential PDF files with content-addressing, and the Ethereum Sepolia blockchain serves as the immutable on-chain record for credential hashes.

The overall system architecture is illustrated in Figure 2.1.

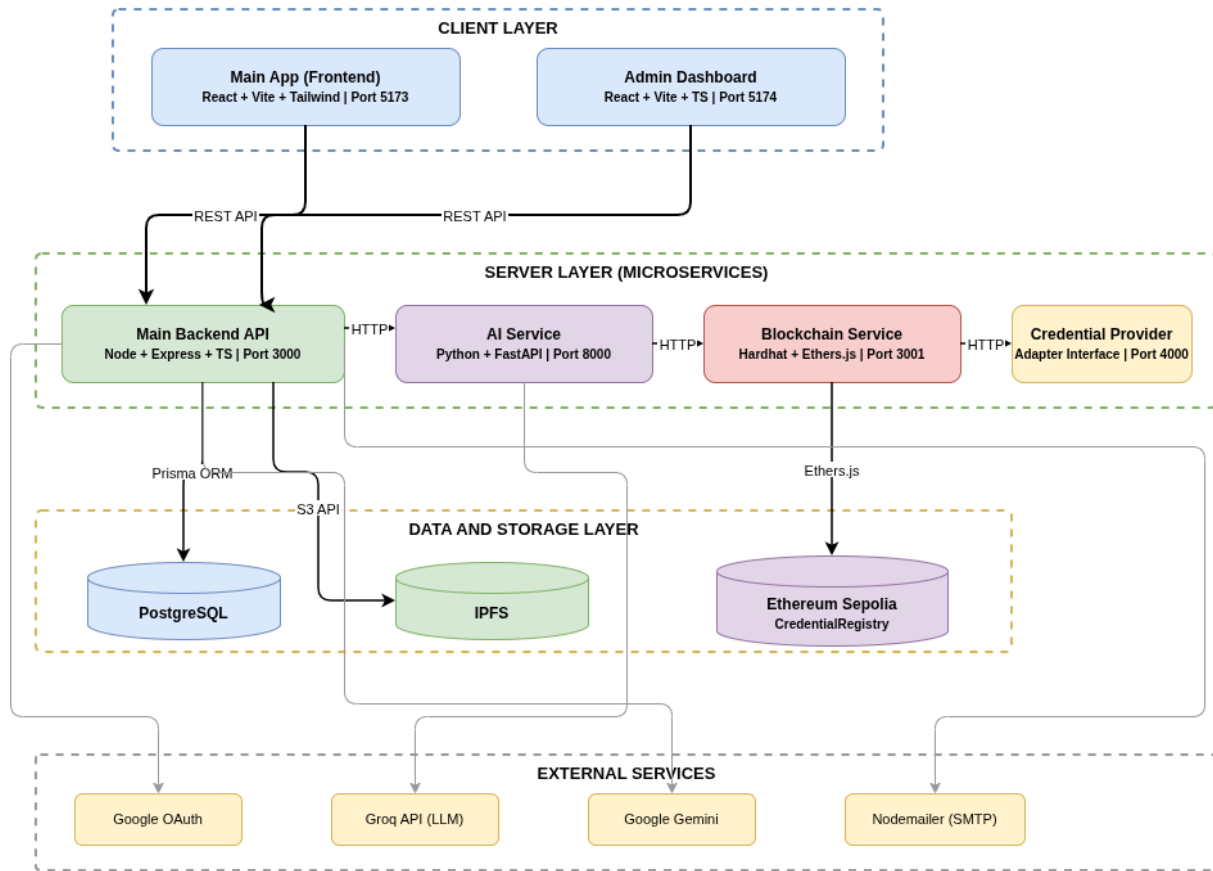


Figure 2.1: System Architecture Overview of MicroMerit Portal

CHAPTER 3: MODELING AND DESIGN

3.1 Introduction

This chapter presents the system design of MicroMerit Portal through standard software engineering diagrams. The design covers the database structure, use cases, data flows, and key process flows including credential issuance, blockchain anchoring, verification, and external sync.

3.2 System Design Overview

The system follows a layered microservices architecture with a clear separation between the presentation layer (React frontends), the application layer (Node.js + Python services), and the data layer (PostgreSQL, IPFS, Blockchain). Each microservice exposes a well-defined HTTP interface and can be independently deployed and scaled.

The design ensures that no user-facing HTTP request is blocked by long-running operations. Credential anchoring to the blockchain and IPFS upload are handled asynchronously via background workers (setImmediate), allowing the HTTP response to return immediately with a “pending” status while the on-chain operations complete in the background.

3.3 Entity Relationship Diagram

The database schema consists of ten entities. The central entity is the Credential table, which holds a unique credential_id (UUID v4), the SHA-256 data_hash, IPFS CID, blockchain transaction hash, and rich metadata including sector, occupation, and tags. Each credential belongs to one Issuer and optionally one Learner (nullable, to support pending-claim credentials issued before the learner registers).

The BulkUploadBatch entity tracks progress of bulk issuance jobs, with a one-to-many relationship to both Credential (the issued records) and BulkUploadError (records of individual file failures). The verification_session entity manages OTP sessions for all user roles independently. Table 3.1 summarizes all entities and their primary keys.

Table 3.1: Database Entities

Entity	Primary Key	Description
issuer	id (INT)	Registered credential-issuing institution
learner	id (INT)	Student/credential holder
admin	id (INT)	Platform administrator
employer	id (INT)	Organization searching/verifying credentials
Credential	id (UUID)	Issued credential with blockchain and IPFS references
issuer_api_key	id (INT)	API key for programmatic credential issuance
BulkUploadBatch	id (INT)	Batch job tracking for bulk issuance
BulkUploadError	id (INT)	Individual file errors within a batch
employer_activity_log	id (INT)	Audit log of employer credential access
verification_session	id (UUID)	OTP verification session (all roles)

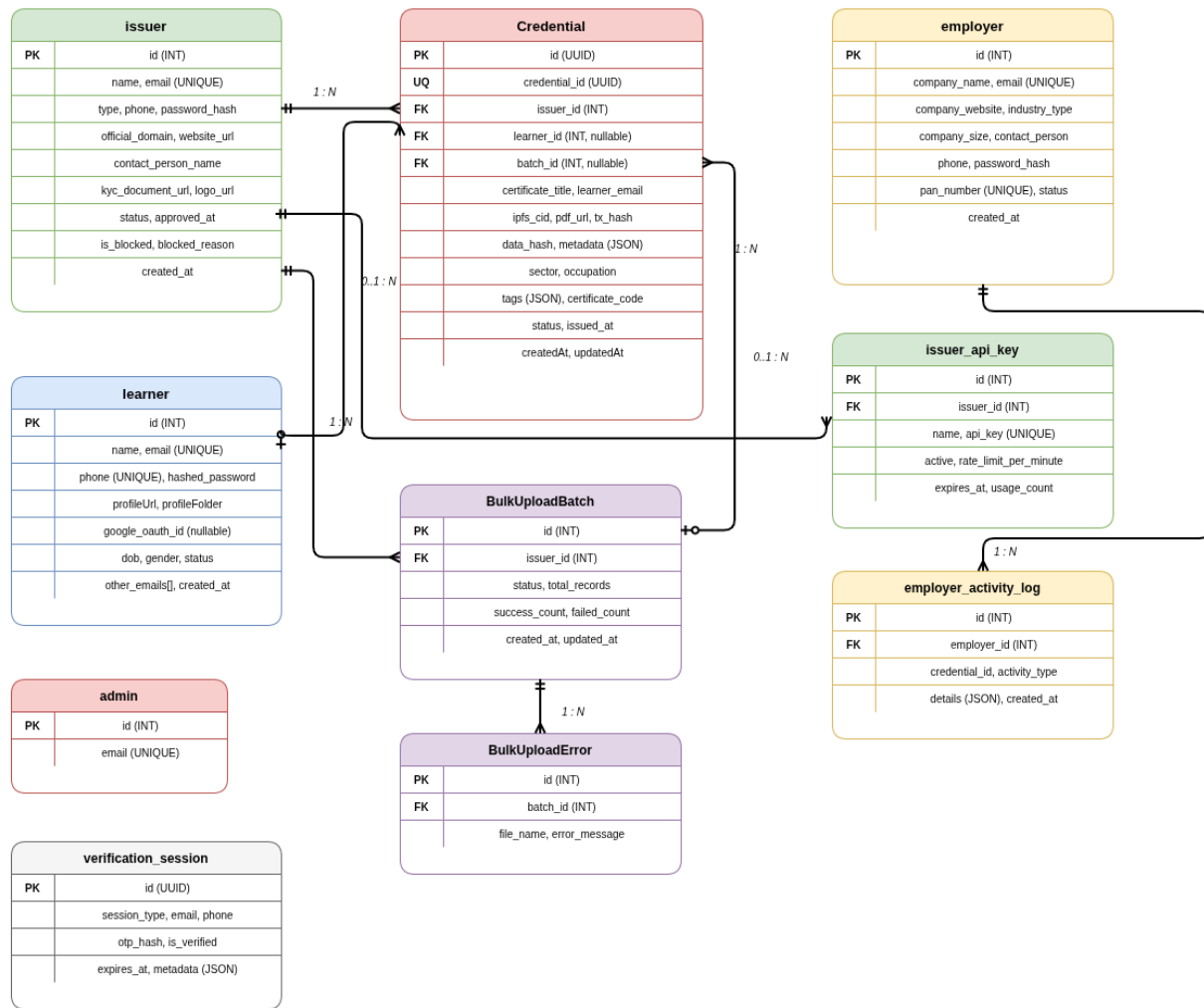


Figure 3.1: Entity Relationship Diagram

3.4 Use Case Diagram

The system supports four distinct actors. The Learner interacts primarily with the credential dashboard, external sync, and public profile features. The Issuer interacts with credential issuance (single and bulk) and API key management. The Employer accesses candidate search, verification, AI chatbot, and comparison features. The Admin operates through a separate portal for issuer governance and platform monitoring. Several system-level use cases (Blockchain Anchoring, IPFS Storage, AI OCR Extraction) are included as dashed «include» relationships to show their automatic invocation.



Figure 3.2: Use Case Diagram

3.5 Data Flow Diagram – Level 0

The Level 0 DFD (Context Diagram) presents the system as a single process of receiving data from five external entities: Learner, Issuer, Employer, Admin, and External Credential Providers. Outward data flows go to the Ethereum Blockchain (data_hash, credential_id), IPFS (PDF files), and back to users (verification results, credential data, search results).

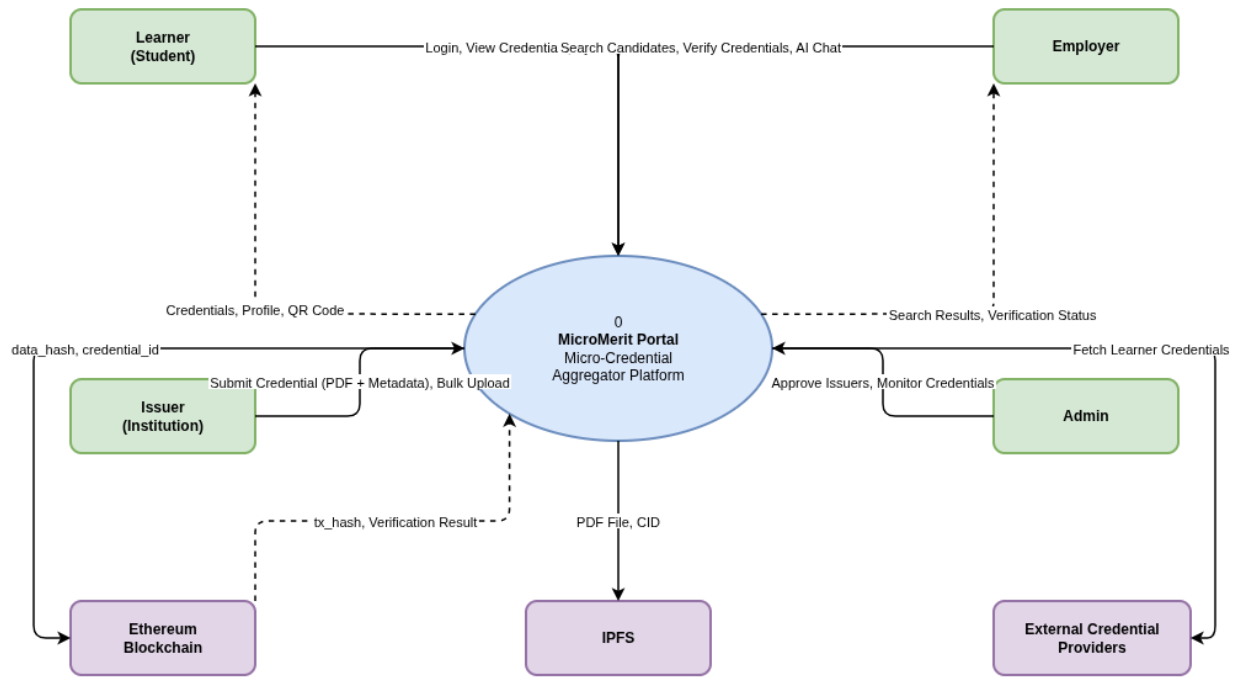


Figure 3.3: DFD Level 0 – Context Diagram

3.6 Data Flow Diagram – Level 1

The Level 1 DFD decomposes the system into seven processes: (1.0) Authentication, (2.0) Credential Issuance, (3.0) Blockchain Anchoring, (4.0) Verification, (5.0) External Sync, (6.0) Employer Search + AI, and (7.0) Admin Management. Three data stores are shown: D1 – PostgreSQL Database, D2 – IPFS Storage, and D3 – Ethereum Blockchain. Processes 3.0 and 4.0 interact directly with D2 and D3, while all other processes primarily interact with D1.

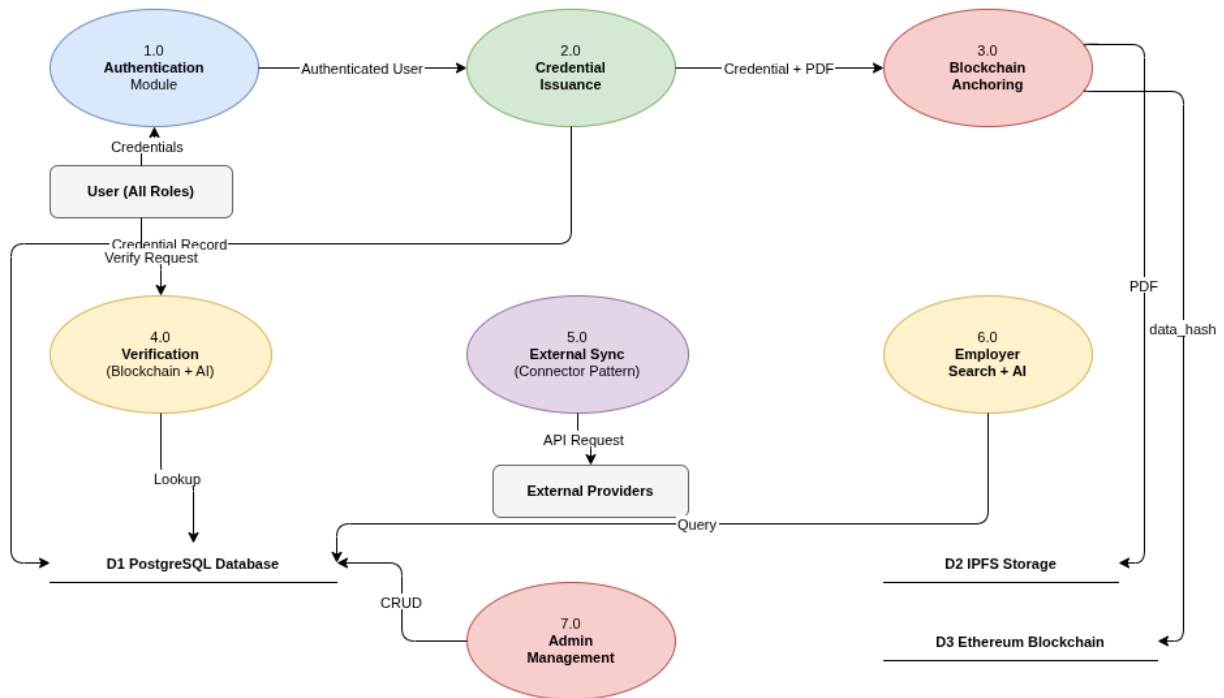


Figure 3.4: DFD Level 1

3.7 Credential Issuance and Blockchain Anchoring Flow

When an issuer submits a credential, the system follows a seven-step pipeline:

- Generate a UUID v4 as the `credential_id`.
- Embed `credential_id` into the PDF's Keywords metadata field using the pdf-lib library.
- Compute SHA-256 of the modified PDF and store it as checksum for PDF-based tamper detection.
- Construct a canonical JSON representation using fixed fields (`credential_id`, `learner_id`, `learner_email`, `issuer_id`, `certificate_title`, `issued_at`, etc.).
- Set `ipfs_cid` and `tx_hash` to null while building the canonical JSON.
- Sort all JSON keys alphabetically at every nesting level before serialization.
- Compute SHA-256 of the canonical JSON string to obtain `data_hash` (anchored on-chain).

After creation, the system persists the credential and completes blockchain/IPFS operations asynchronously:

- Save the credential record to PostgreSQL with `blockchain_status`: pending.
- Return the HTTP response immediately to the issuer (no waiting for confirmations).

- Run two background jobs concurrently via `setImmediate`.
- Job 1: Call the Blockchain Service to invoke `issueCredential()` on Sepolia and record the returned `tx_hash`.
- Job 2: Upload the PDF to IPFS and record the returned `ipfs_cid`.
- Write both `tx_hash` and `ipfs_cid` back to the database upon completion.

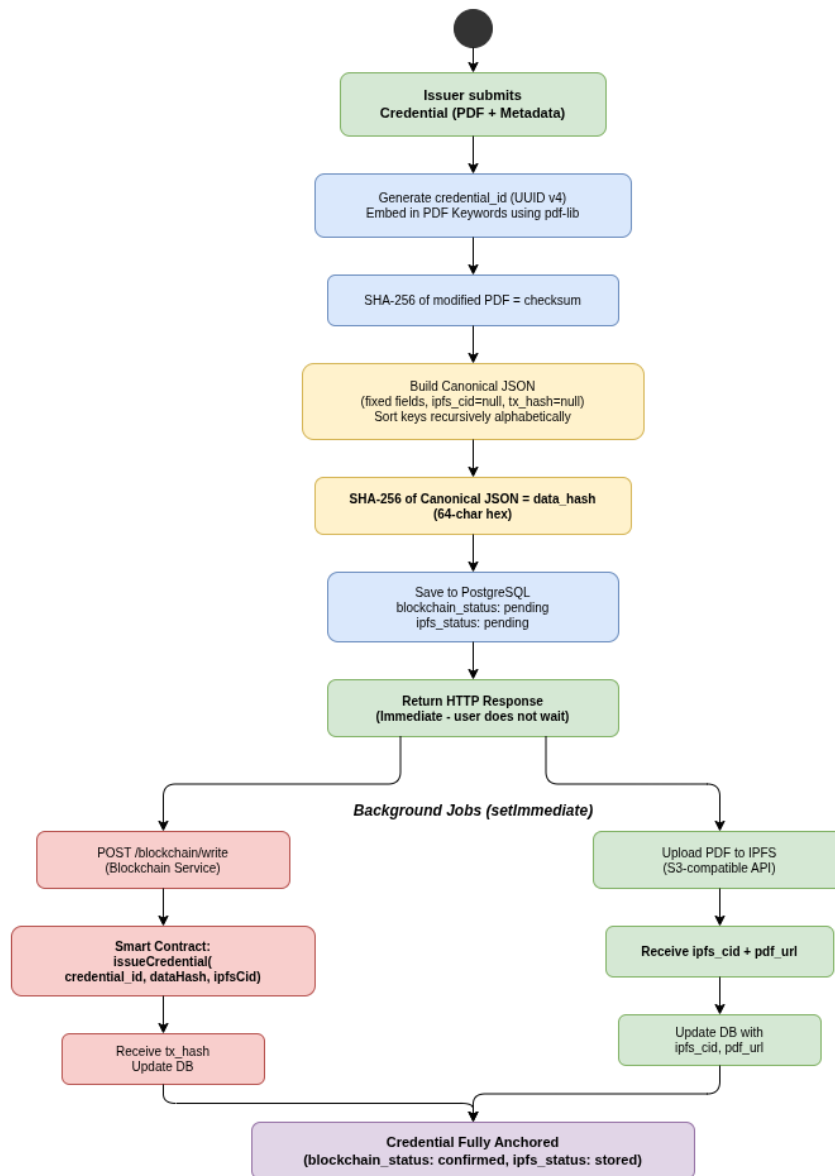


Figure 3.5: Credential Issuance and Blockchain Anchoring Flow

3.8 Credential Verification Flow

The platform implements two verification methods because they serve complementary purposes:

- Blockchain verification is cryptographically absolute, but typically requires the exact original credential identifier or file.
- Scanned photocopies or printed certificates change bytes and therefore fail byte-level hashing even when genuine.
- AI verification fills this gap by performing semantic content comparison rather than byte-level comparison.

For blockchain verification, the verifier may use either identifier-based verification (Sub-method A) or PDF-based verification (Sub-method B):

- **Sub-method A (by credential_id / tx_hash / IPFS CID):** Reconstruct the canonical JSON from the database (with ipfs_cid and tx_hash set to null), recompute the SHA-256 hash, compare it with stored data_hash, and query the CredentialRegistry smart contract on Sepolia. Both checks must pass to return VALID.
- **Sub-method B (by original PDF upload):** Extract credential_id from the PDF's Keywords field, compute SHA-256 of the uploaded file, and compare it with the stored checksum. Any byte-level modification results in INVALID.

For AI verification, the system performs probabilistic document comparison:

- Verifier provides a credential_id and uploads any document copy (photo, printout, or scan).
- System retrieves the original PDF from IPFS and sends both documents to Google Gemini 2.5 Flash for field-by-field comparison.
- Gemini returns match verdict, confidence score (0–100%), mismatched fields, and a summary sentence.
- UI labels AI verification as probabilistic and recommends blockchain verification for legal proof.

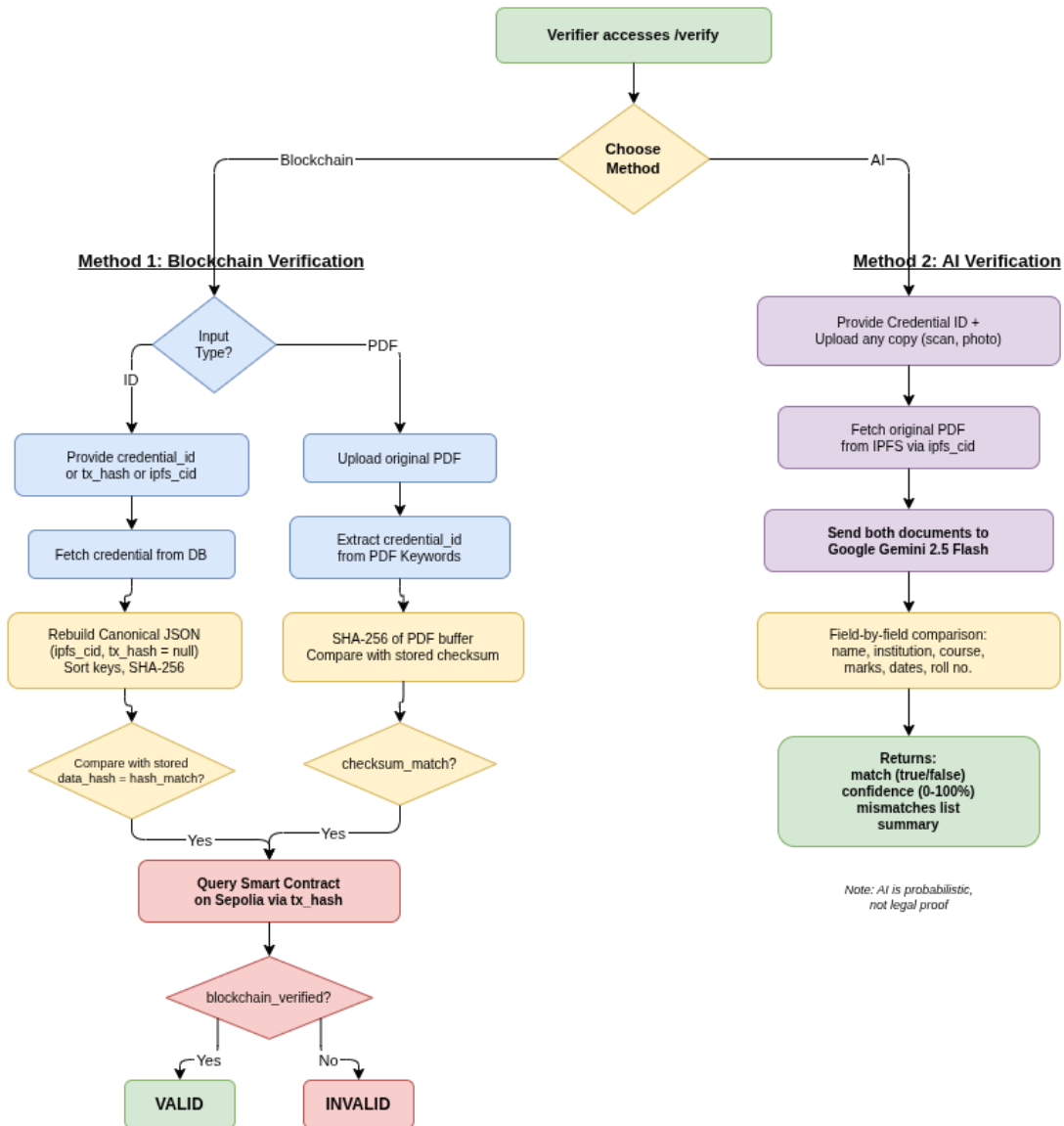


Figure 3.6: Dual-Method Credential Verification Flow

3.9 External Credential Sync Design

External credential sync uses a provider-agnostic connector pattern. The ConnectorFactory selects the appropriate connector (Udemy, Credly, etc.) based on the provider identifier in the sync request. Each connector implements a common interface with methods for authentication, credential fetching, and response parsing. A separate server (the Credential Provider Abstraction

Layer, Port 4000) acts as a mock provider during development, simulating the response formats of real external APIs.

After connector fetches credentials from the external provider, the raw response is normalized to the platform's standard Credential schema. If the learner's email is already registered in the system, the normalized credential is directly linked to their account. If the learner does not yet have an account, a pending-claim record is created, and an email invitation is sent. The pending credential is resolved when the learner registers with the matching email.

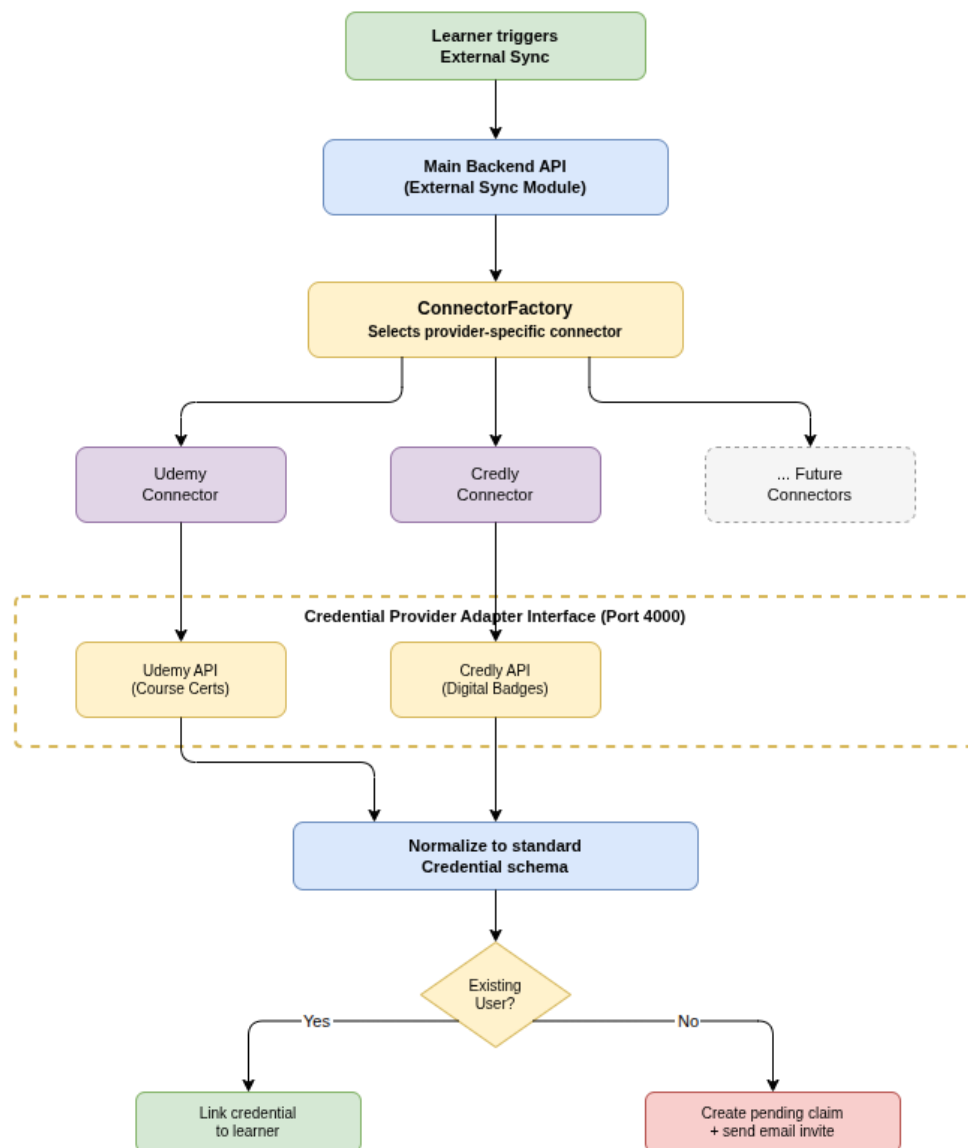


Figure 3.7: External Credential Sync – Connector Pattern

3.10 Bulk Issuance Adapter Pattern

Bulk issuance is designed around the adapter pattern. The `IssuerBulkAdapterFactory` receives each file extracted from the uploaded ZIP and selects the appropriate adapter. Each adapter implements four interface methods: `identify()` to detect the file type, `extractMetadata()` to parse the credential data (either directly from JSON or via AI OCR for PDF/image files), `verify()` to validate the parsed data against required fields, and `normalize()` to map the data to the platform's standard Credential schema. After normalization, each credential goes through the standard blockchain anchoring pipeline. The `BulkUploadBatch` record tracks total, success, and failed counts, which the frontend polls to show a real-time progress indicator.

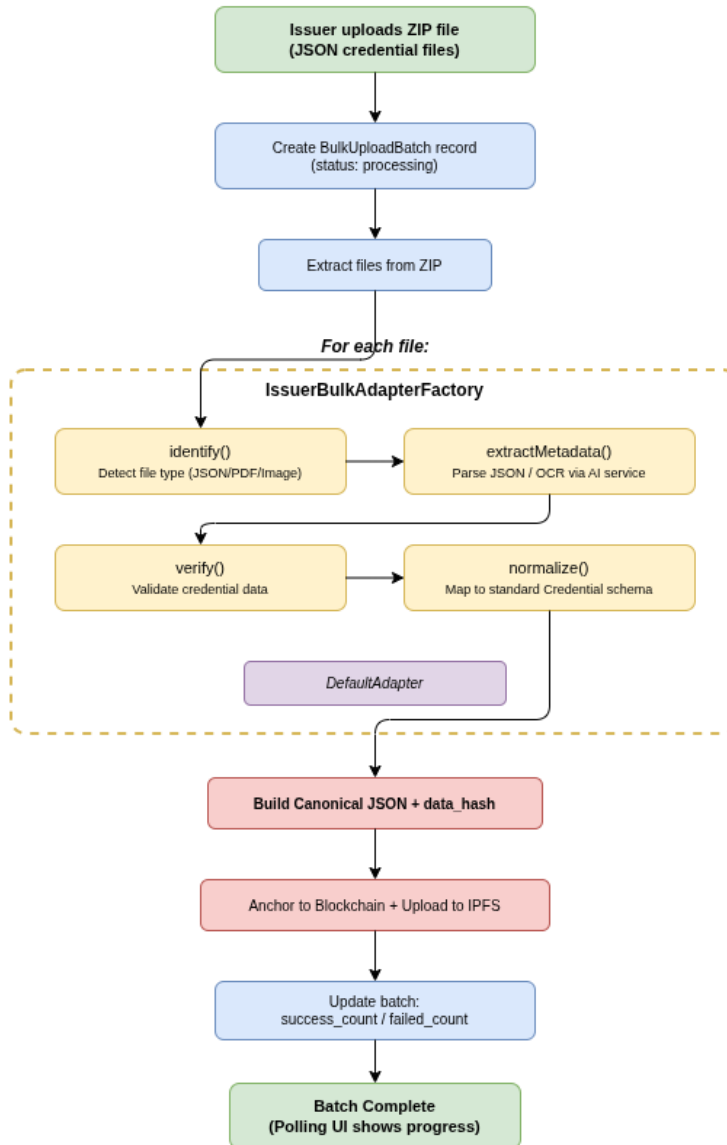


Figure 3.8: Bulk Issuance – Adapter Pattern Flow

3.11 Authentication Flow

The platform supports two authentication methods for learners: Email + OTP and Google OAuth 2.0. Issuers and employers use Email + OTP only. The Email + OTP flow involves submitting credentials, verifying them with bcrypt, sending an OTP via Nodemailer, and confirming the OTP against its stored hash. Google OAuth follows the standard authorization code flow, with the backend finding or creating a user account upon successful callback. Regardless of the

authentication method, all paths converge at JWT token generation — the token encodes the user's role, ID, and email, and is stored in the browser via Redux Persist.

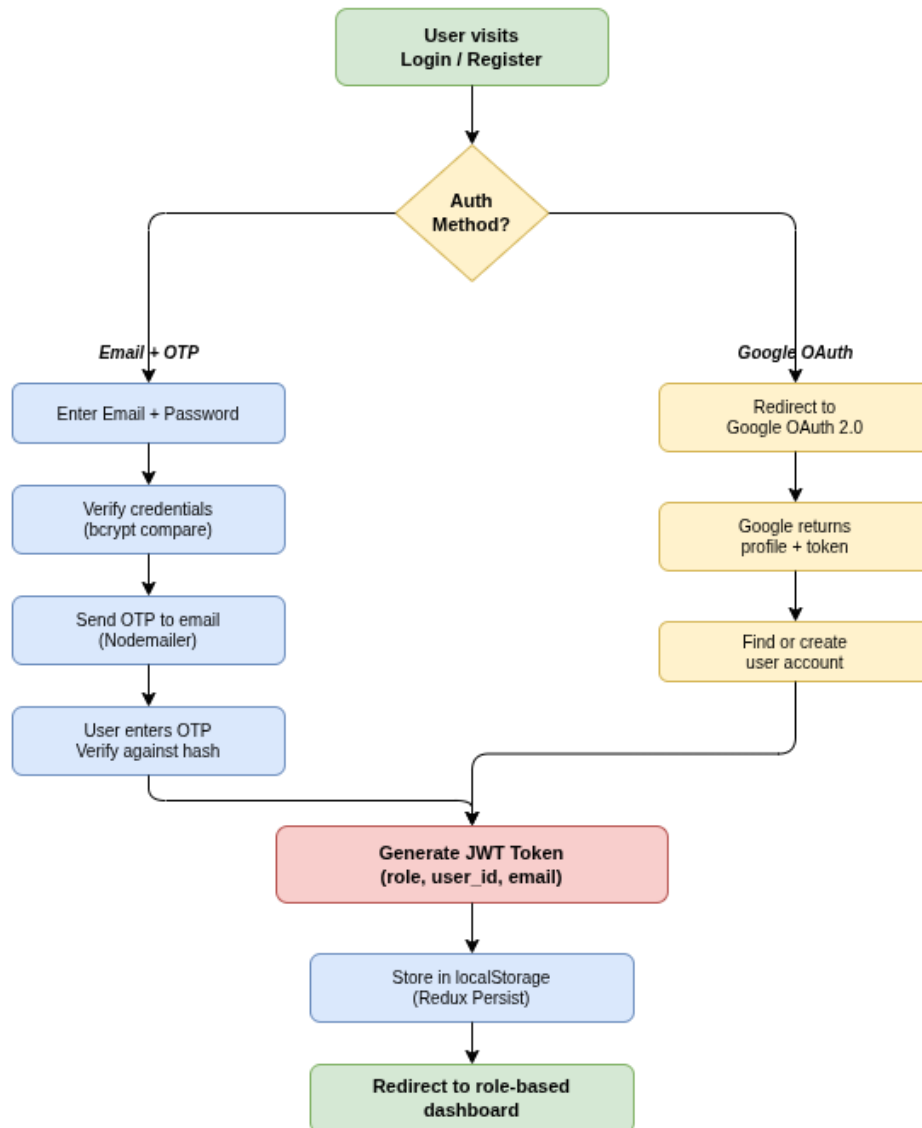


Figure 3.9: Multi-Method Authentication Flow

CHAPTER 4: IMPLEMENTATION

4.1 Introduction

This chapter describes the implementation of MicroMerit Portal, covering the six microservices, module-wise functionality, API design, and user interface. The system was implemented using the technologies described in Chapter 2, and all features described in the design chapter have been realized in working software.

4.2 Module-wise Implementation

4.2.1 Authentication Module

The authentication module handles registration and login for all four user roles:

- Learners authenticate via Email + OTP or Google OAuth 2.0 (custom callback route creates or finds the account).
- OTPs are hashed before storage in the `verification_session` table and expire after a configurable duration.
- On success, the backend returns a signed JWT containing role, ID, and email.

4.2.2 Credential Issuance Module

Credential issuance is the core module of the platform and is implemented as follows:

- Issuer submits a PDF and metadata (learner email, certificate title, sector, occupation, tags) through the web portal.
- Backend embeds a UUID v4 `credential_id` into the PDF's Keywords metadata using `pdf-lib`.
- System computes a SHA-256 checksum, builds canonical JSON, and stores the credential with pending status.
- HTTP response is returned immediately while background workers complete blockchain anchoring and IPFS upload in parallel.
- Issuers can also issue credentials programmatically via REST API using an API key.

4.2.3 Bulk Issuance Module

The bulk issuance module supports high-volume issuance through ZIP uploads:

- Issuer uploads a ZIP archive containing JSON files (one credential per file).
- Server extracts the archive, creates a BulkUploadBatch record, and processes files through IssuerBulkAdapterFactory.
- Each credential runs independently through embedding, hashing, blockchain anchoring, and IPFS upload.
- Issuer dashboard polls the batch endpoint to render total, success, and failed counts in real time.

4.2.4 Credential Verification Module

The verification module exposes public endpoints and supports multiple verification paths:

- Blockchain verification by identifier: reconstruct canonical JSON, recompute data_hash, compare with stored value, and query the Blockchain Service for on-chain confirmation.
- PDF-based verification: compute SHA-256 checksum of uploaded PDF and compare it with the stored checksum.
- AI verification: retrieve original PDF from IPFS and forward both documents to Google Gemini 2.5 Flash for comparison.

4.2.5 External Credential Sync Module

The external sync module uses the connector pattern to fetch credentials from external providers.

The ConnectorFactory maintains a registry of provider-specific connectors, each handling authentication, request formatting, and response parsing for their respective API. Fetched credentials are normalized to the platform schema and linked to learner accounts, with a pending-claim mechanism for learners who have not yet registered.

4.2.6 Employer Module

The employer module provides three features. The candidate search interface supports filtering by skill keyword, certificate title, issuer name, learner name, and location, returning paginated results that can be exported to CSV. The AI chatbot allows an employer to enter a learner's email and a natural-language question; the backend fetches all verified credentials for that learner and sends them as structured context to the Groq-hosted Llama 3.1 8B model, which returns an AI-

generated answer with referenced certificate titles and a confidence score. The candidate comparison feature allows selecting up to three learners and generating a side-by-side AI comparison.

4.2.7 Admin Module

The admin module operates through a separate React portal (Port 5174) and is accessible only to admin-role accounts. Admins can review issuer registration requests and approve or reject them (with a rejection reason stored on the issuer record). The admin dashboard also provides credential monitoring across all issuers and controls for enabling or disabling external credential sync per provider.

4.2.8 Blockchain Service

The Blockchain Service is an independent TypeScript microservice (Port 3001) that wraps Ethers.js interactions with the CredentialRegistry smart contract. It exposes two endpoints: POST /blockchain/write (to call issueCredential() on-chain and return the tx_hash) and GET /blockchain/verify (to call verifyCredential() and getCredential() on-chain and return the result). In development and demo mode, BLOCKCHAIN MOCK_ENABLED=true generates a deterministic fake tx_hash without actually broadcasting to Sepolia, while IPFS uploads continue to run with real content.

4.2.9 AI Service

The AI service (Port 8000) is a Python FastAPI application that exposes an OCR extraction endpoint. An uploaded PDF or image is processed by Tesseract OCR to extract raw text, which is then sent to the Groq API with a structured prompt instructing the Llama 3.1 8B model to extract specific fields: learner name, institution, course title, issue date, expiry date, and sector. The service returns a structured JSON response that the main backend uses to pre-fill the issuance form.

4.3 API Design

The main backend exposes RESTful endpoints for authentication, credential management, verification, employer workflows, and administrative governance. Table 4.1 summarizes the key API endpoints used across the platform.

Table 4.1: Key API Endpoints

Method	Endpoint	Role	Description
POST	/auth/learner/register	Public	Learner registration
POST	/auth/learner/verify-otp	Public	OTP verification, returns JWT
GET	/auth/google/callback	Public	Google OAuth callback
POST	/credentials/issue	Issuer	Issue single credential
POST	/credentials/bulk-upload	Issuer	Upload ZIP for bulk issuance
GET	/credentials/:id	Auth	Get credential details
POST	/credentials/verify	Public	Blockchain-based verification
POST	/credentials/verify-ai	Public	AI document comparison verification
POST	/credentials/verify-pdf	Public	PDF checksum verification
GET	/learner/profile	Learner	Get own profile and credentials
GET	/p/:slug	Public	Public learner profile page data
GET	/employer/search	Employer	Search learners by filters
POST	/employer/chat	Employer	AI chatbot query
GET	/admin/issuers	Admin	List all issuers pending approval
POST	/admin/issuers/:id/approve	Admin	Approve issuer registration

All endpoints return JSON responses with a consistent structure: {success: boolean, data: object, message: string}. Error responses include an appropriate HTTP status code.

4.4 User Interface Implementation

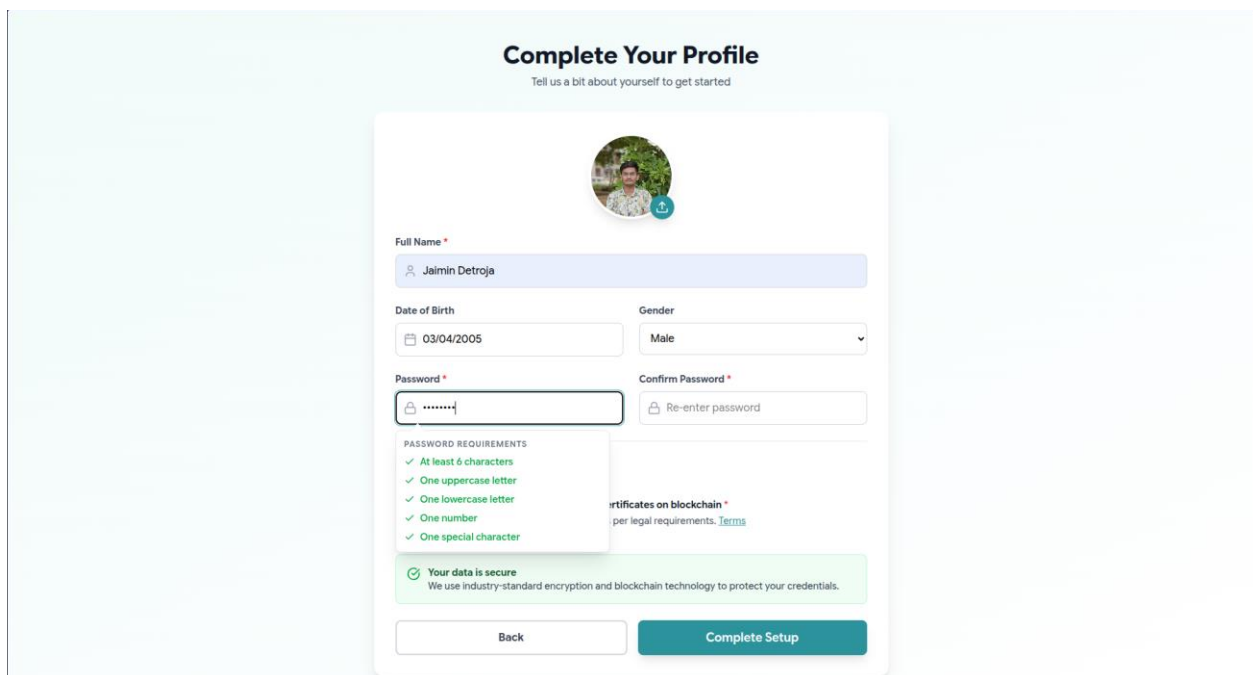
The frontend is implemented in React 18 with Vite and Tailwind CSS, providing responsive, role-specific interfaces for Learners, Issuers, Employers, and Administrators. Each role has a dedicated section of the application with tailored navigation and functionality. The following subsections describe key screens across all four roles, accompanied by screenshots.

4.4.1 Learner Interface

The Learner role has six key screens covering the full credential lifecycle from account creation to credential management and public profile sharing.

Signup — Step 3 (Role Selection & Profile Completion)

The third step of the signup flow prompts the learner to select their role (Learner, Issuer, or Employer), enter their Student ID, and set their profile slug for the public profile URL. This step completes account creation and redirects to the role-specific dashboard.



The screenshot displays a web form titled "Complete Your Profile" with the subtitle "Tell us a bit about yourself to get started". The form is set against a light blue background. At the top, there is a circular profile picture placeholder with a blue upload icon. Below this, the form fields are organized as follows: a "Full Name" field with a red asterisk and a search icon, containing the text "Jalmin Detroja"; a "Date of Birth" field with a calendar icon, containing "03/04/2005"; a "Gender" dropdown menu currently set to "Male"; a "Password" field with a red asterisk, a lock icon, and masked characters "....."; and a "Confirm Password" field with a red asterisk and the placeholder text "Re-enter password". A "PASSWORD REQUIREMENTS" section lists five criteria, each with a green checkmark: "At least 6 characters", "One uppercase letter", "One lowercase letter", "One number", and "One special character". To the right of these requirements, there is a note about "Certificates on blockchain" with a link to "Terms". At the bottom of the form, a green banner states "Your data is secure" with a checkmark icon and the text "We use industry-standard encryption and blockchain technology to protect your credentials." Below the banner are two buttons: a "Back" button and a teal "Complete Setup" button.

Figure 4.1: Learner Registration — Step 3 (Role & Profile Setup)

Learner Dashboard

The learner dashboard presents a summary card showing the total number of verified credentials, blockchain-confirmed credentials, and pending credentials. Below the summary, a credential grid displays each certificate with its title, issuing institution, issue date, and blockchain status badge (Pending / Confirmed).

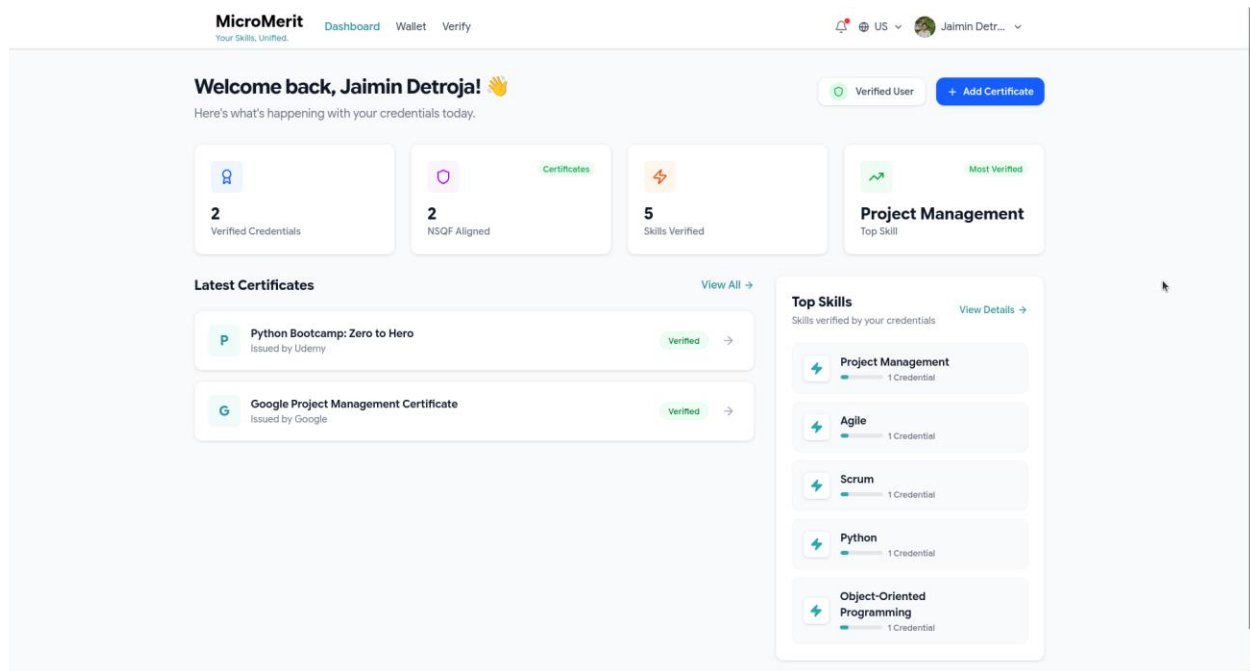


Figure 4.2: Learner Dashboard — Credential Overview

Credential Wallet

The wallet page lists all credentials belonging to the learner in a detailed table or card view, including IPFS CID, transaction hash, and a download button for the original PDF. Learners can filter credentials by issuer, date range, or blockchain status.

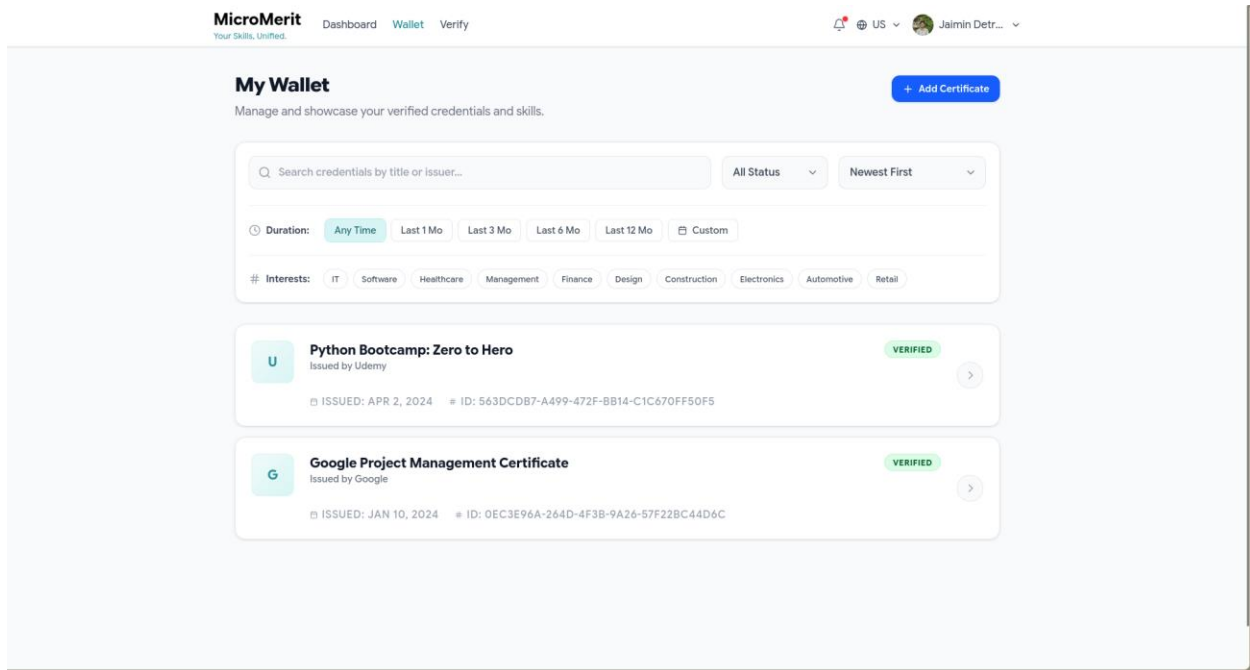


Figure 4.3: Learner Credential Wallet

Individual Credential View

Clicking any credential opens its detail page, showing the full certificate metadata, an embedded PDF viewer for the original document stored on IPFS, the blockchain transaction hash linked to Etherscan, and a QR code that links to the public credential verification page.

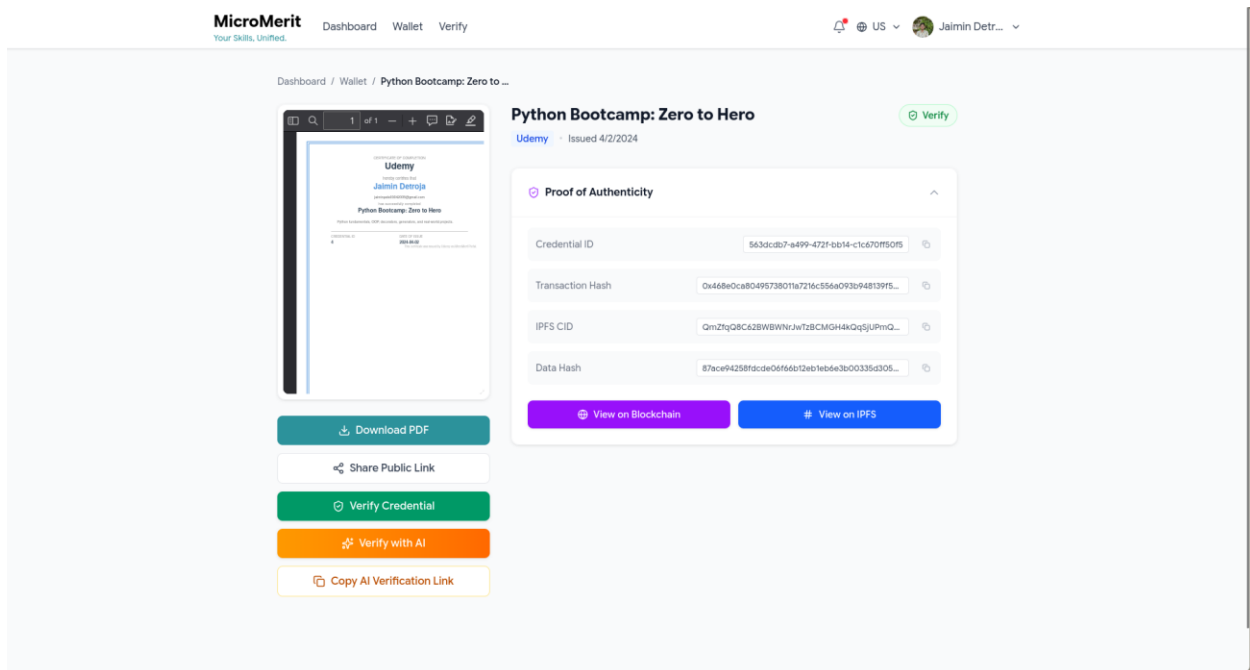


Figure 4.4: Individual Credential Detail View

Add Certificate (External Credential Sync)

The Add Certificate page allows learners to link credentials from external providers (NSDC, Udemy, Credly-like platforms). The learner selects the provider, enters their provider account identifier, and the system fetches and normalizes their credentials through the connector pattern, adding them to the wallet.

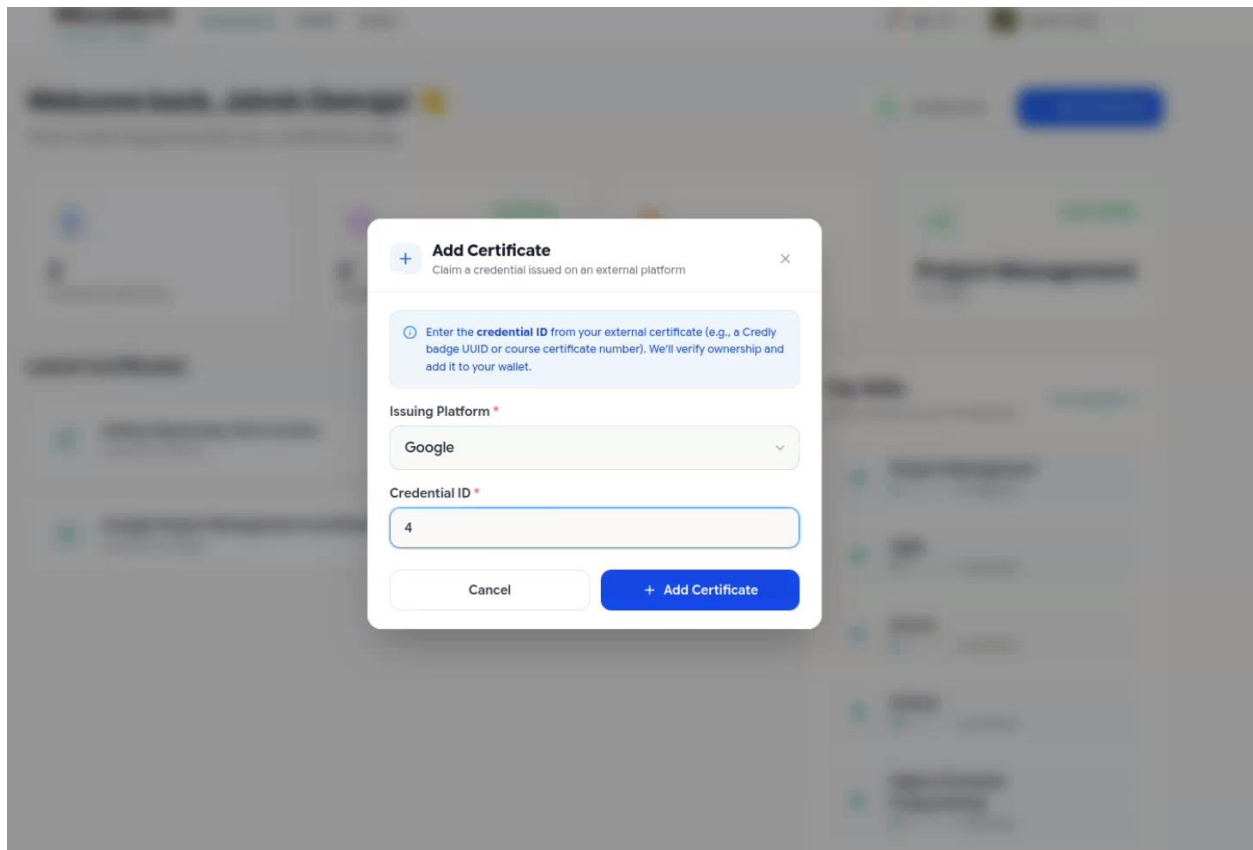


Figure 4.5: External Credential Sync — Add Certificate

Learner Profile

The profile page allows the learner to update personal information, upload a profile photo, and manage their public profile slug. A preview link shows how their public profile page (/p/:slug) appears to employers and verifiers.

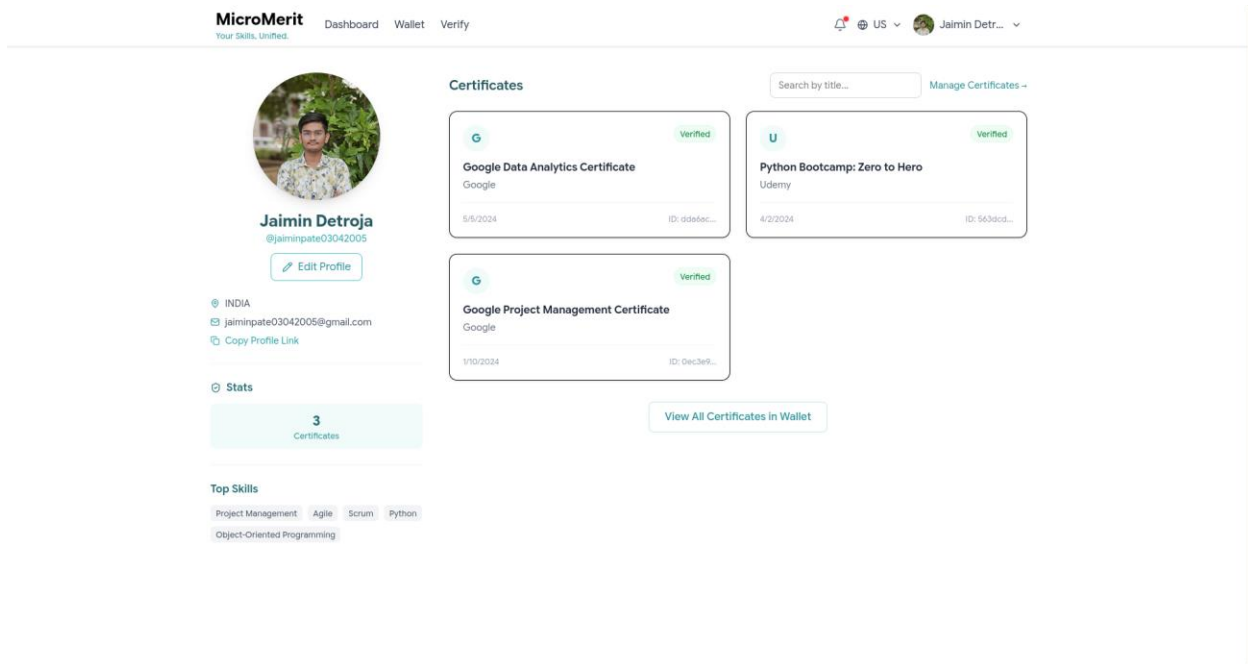


Figure 4.6: Learner Profile Management

4.4.2 Issuer Interface

The Issuer role provides five functional screens for credential management, analytics, and API integration.

Issuer Dashboard

The issuer dashboard displays summary statistics: total credentials issued, credentials pending blockchain confirmation, confirmed credentials, and failed issuances. Recent issuance activity is listed in a timeline, and quick-action buttons link to the issuance form and bulk upload.

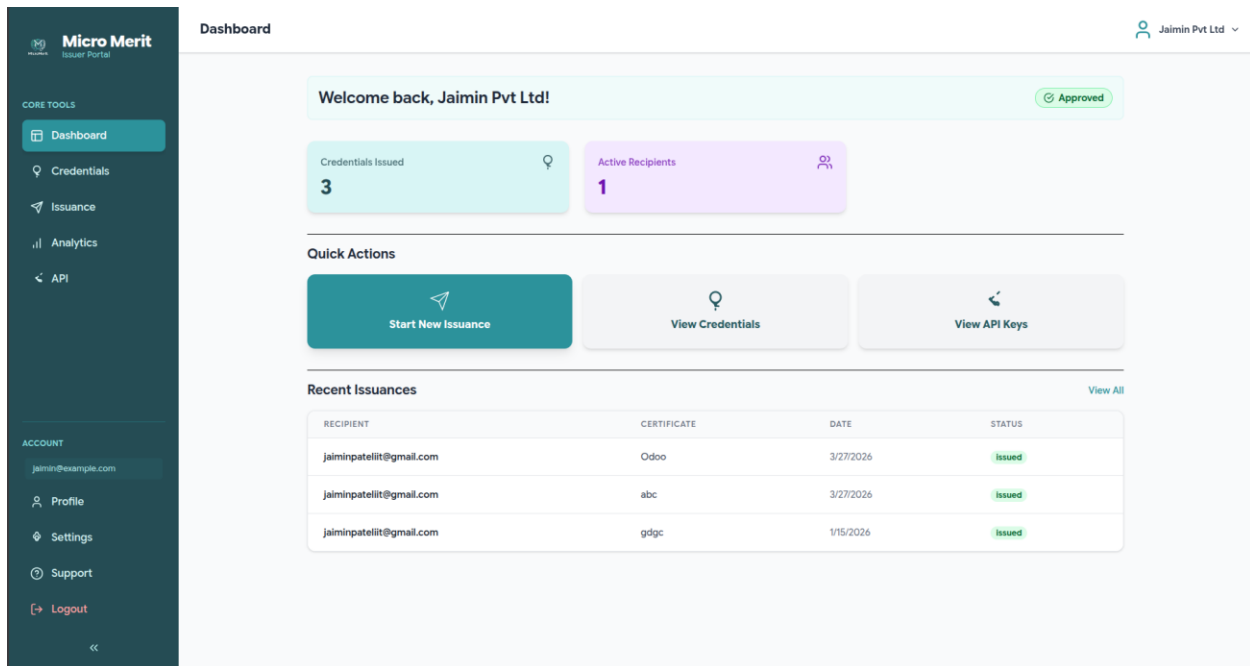


Figure 4.7: Issuer Dashboard — Issuance Overview

Credentials List

The credentials page lists all credentials issued by this issuer with their blockchain status, learner email, certificate title, and issue date. Issuers can search and filter the list, and click into individual credentials to view full details.

CREDENTIAL ID	RECIPIENT	CERTIFICATE TITLE	ISSUED DATE	STATUS	NSQF	ACTIONS
0db1d2f0...	jaiminpatel1@gmail.com	Odoo	3/27/2026	Issued	N/A	View PDF Download QR Verify
410d5ea5...	jaiminpatel1@gmail.com	abc	3/27/2026	Issued	N/A	View PDF Download QR Verify
12fd50ee...	jaiminpatel1@gmail.com	gdgc	1/15/2026	Issued	N/A	View PDF Download QR Verify

Figure 4.8: Issuer Credentials List

Credential Issuance Form

The issuance form requires the issuer to upload the credential PDF and enter the certificate title. On submission, the credential is immediately saved to the database with a Pending blockchain status, and a background job anchors the data hash to the Ethereum Sepolia blockchain and uploads the PDF to IPFS. The credential card appears instantly in the dashboard while anchoring completes asynchronously.

The screenshot displays the 'Micro Merit Issuance Portal' interface. On the left is a dark teal sidebar with the following sections:

- CORE TOOLS**
 - Dashboard
 - Credentials
 - Issuance** (highlighted with a teal background)
 - Analytics
 - API
- ACCOUNT**
 - jaimin@example.com
 - Profile
 - Settings
 - Support
 - Logout

The main content area is titled 'Issuance' and features a 'Start Credential Issuance' section with the subtitle 'Issue secure, blockchain-verified credentials to your learners.' Below this are two buttons: 'Single Issuance' (teal) and 'Bulk Issuance' (white with a teal border). The 'Single Issuance' form is a white box with the following fields:

- Recipient Email**: A text input field containing 'john.doe@email.com'.
- Certificate Title / Course Name**: A text input field containing 'e.g. React Mastery'.
- Certificate File (PDF only)**: A dashed border box containing a file upload icon and the text 'Upload a file or drag and drop' and 'PDF only, up to 10MB'.

At the bottom of the form is a teal button labeled 'Issue Credential' with a right-pointing arrow.

Figure 4.9: Credential Issuance Form

Certificate Added Successfully

This success confirmation screen is displayed immediately after credential submission. It confirms that the certificate has been created, shows the newly added credential card in the issuer's list, and indicates a Pending blockchain status while anchoring completes asynchronously.

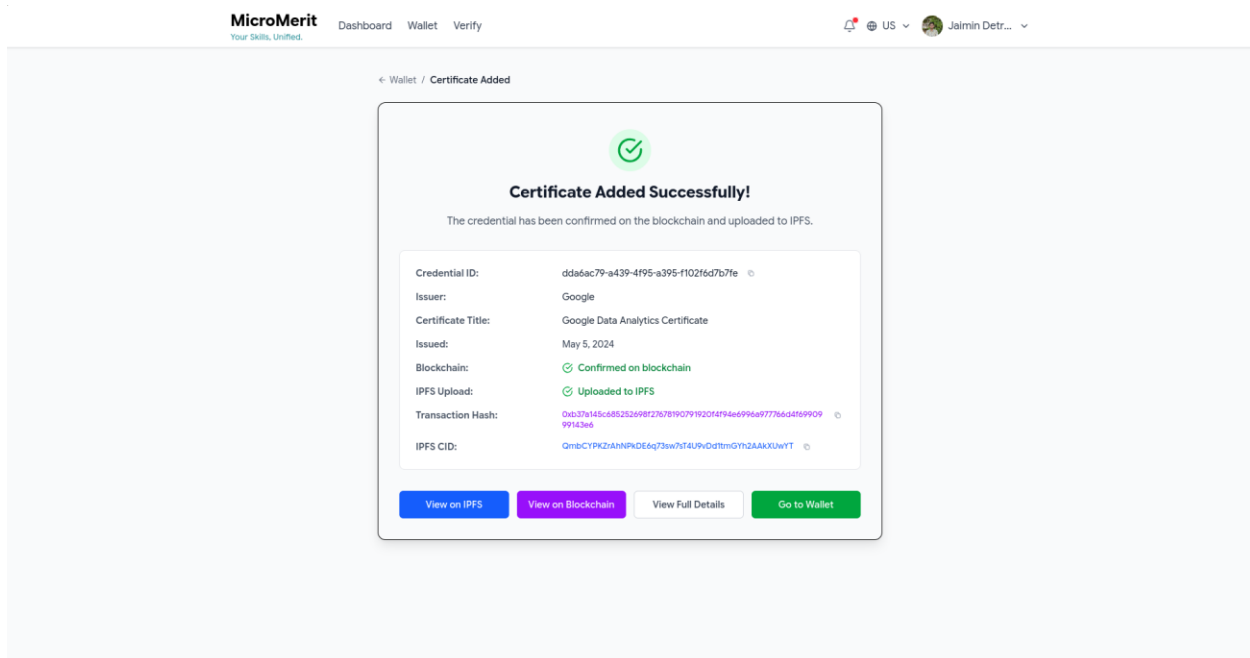


Figure 4.10: Certificate Added Successfully — Pending Blockchain Status

Issued Credential View

After issuance, the issuer can view the credential record showing the embedded PDF, the generated credential ID, the blockchain transaction hash (once confirmed), and the IPFS CID. This screen also displays the learner's name and email and provides a shareable verification link.

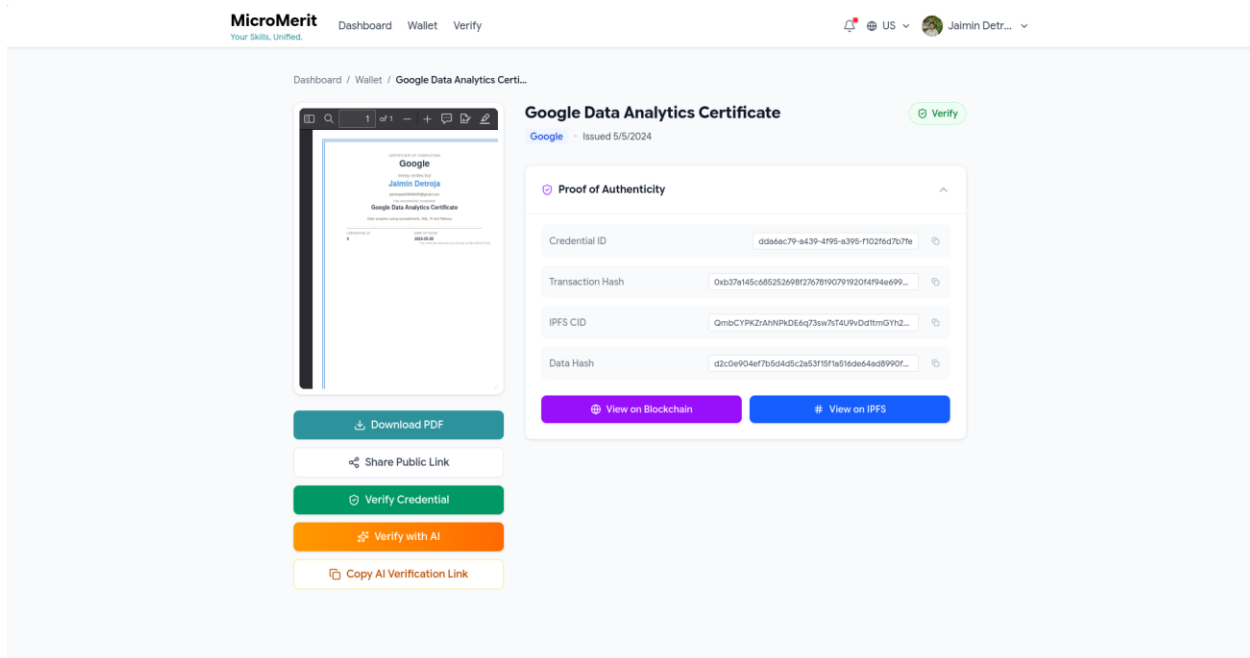


Figure 4.11: Issued Credential Detail View

Analytics

The analytics page provides charts and statistics for the issuer: credentials issued over time (line chart), breakdown by blockchain status (pie chart), and a table of top learners by credential count. This helps issuers monitor issuance volume and blockchain confirmation rates.

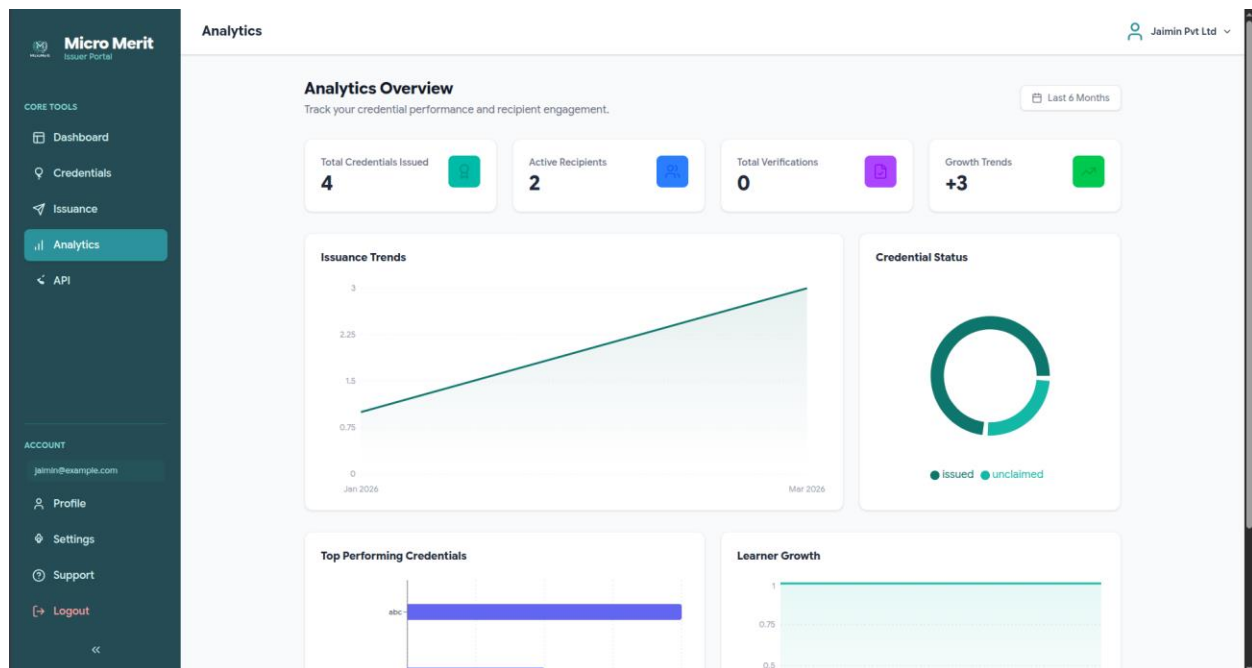


Figure 4.12: Issuer Analytics Dashboard

API Key Management

The API page allows issuers to generate and revoke API keys for programmatic credential issuance. Each key is displayed with its creation date, last-used timestamp, and a revoke button. The page also includes an API usage guide with sample cURL requests for the credential issuance endpoint.

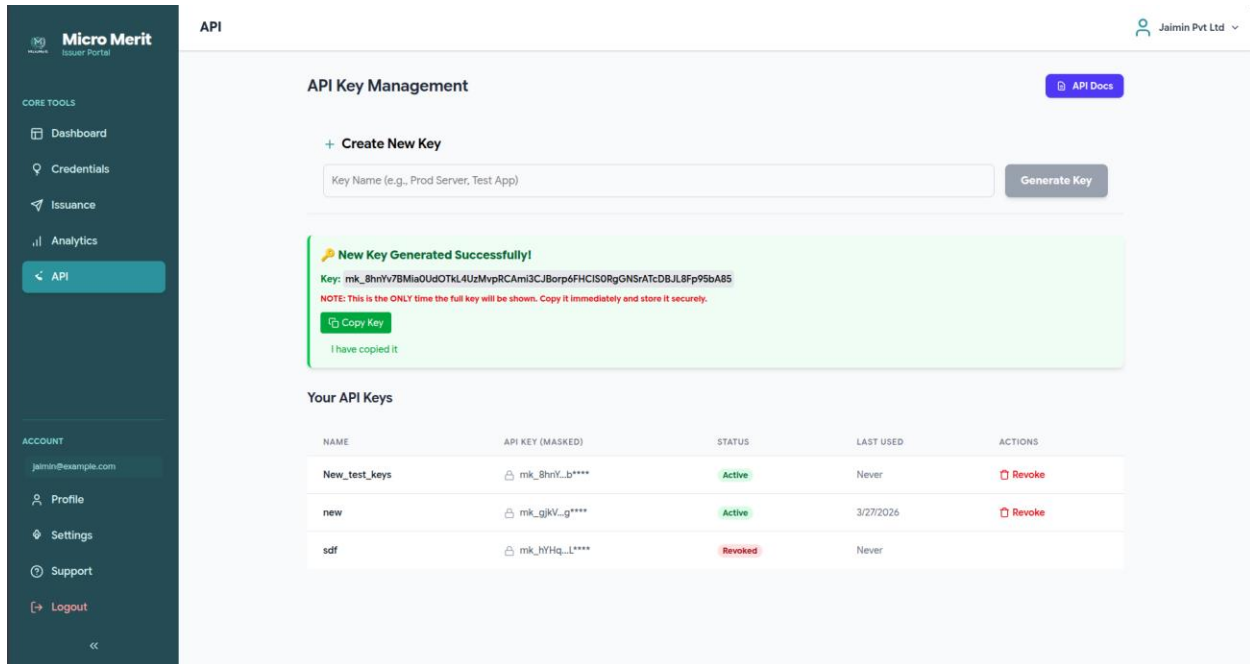


Figure 4.13: Issuer API Key Management

4.4.3 Employer Interface

The Employer role has five screens covering candidate discovery, AI-assisted querying, and credential verification.

Employer Dashboard

The employer dashboard shows high-level activity summary cards including searches performed, credentials verified, and saved profiles. It also provides quick-action links to the Search and Verify pages for fast navigation.

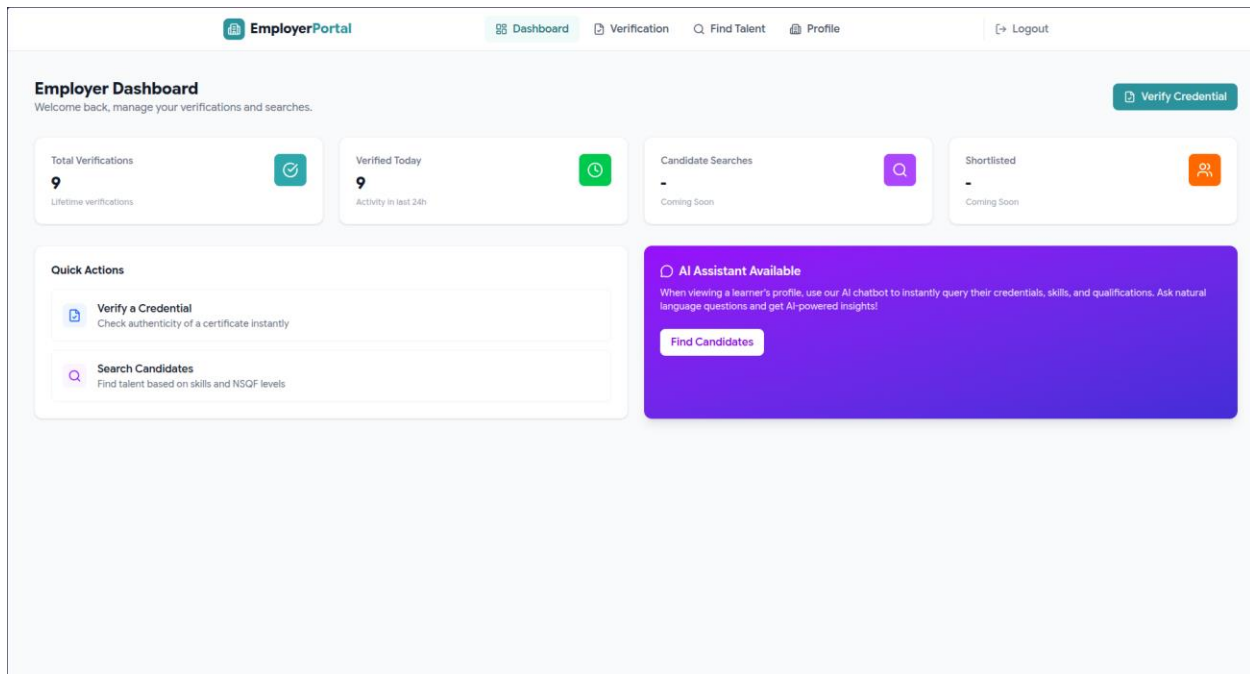


Figure 4.14: Employer Dashboard — Summary and Quick Actions

Candidate Search (/employer/search)

The search page presents a filter panel (skills, certificate title, issuer name, learner name, and location) and a results grid showing matching learner cards with their top credentials and skill tags. Employers can select up to three candidates for an AI-powered side-by-side comparison, and export results as a CSV file.

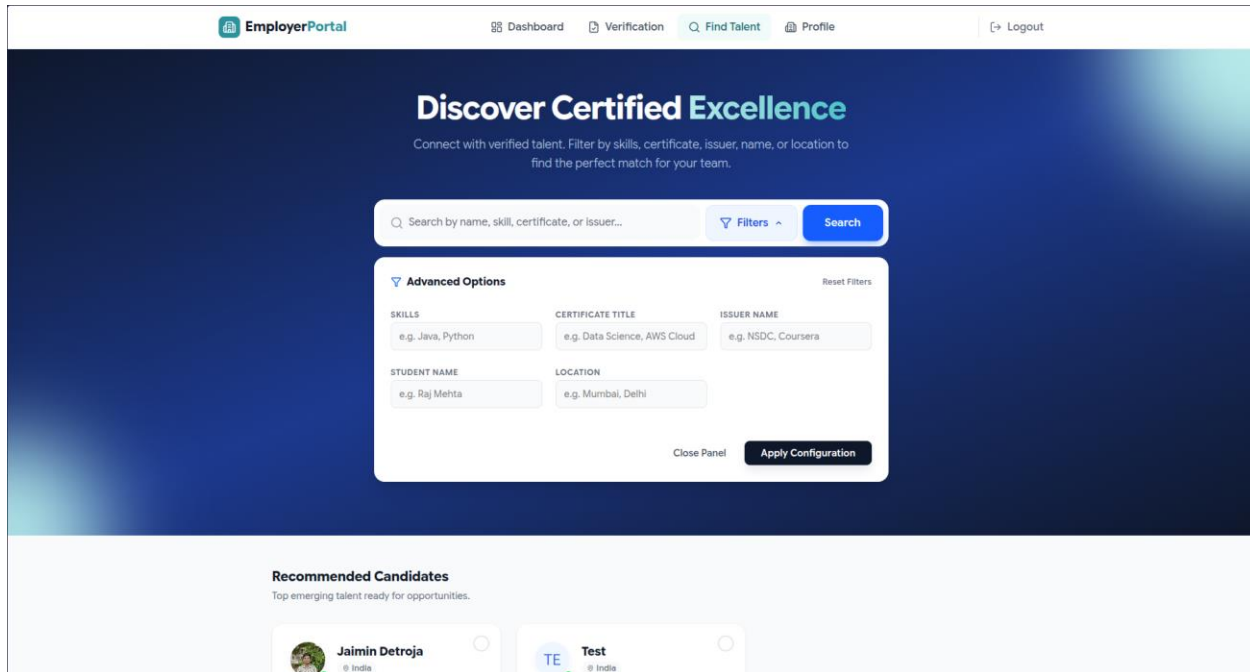


Figure 4.15: Employer Candidate Search

AI Chatbot (on Search Page)

The AI chatbot panel on the search page allows the employer to enter a learner's email and ask a natural-language question (e.g., "Does this candidate have any cloud certifications?"). The backend fetches all verified credentials for that learner and sends them as context to the LLM, which returns an answer with referenced certificates, relevant skills, and a confidence score.

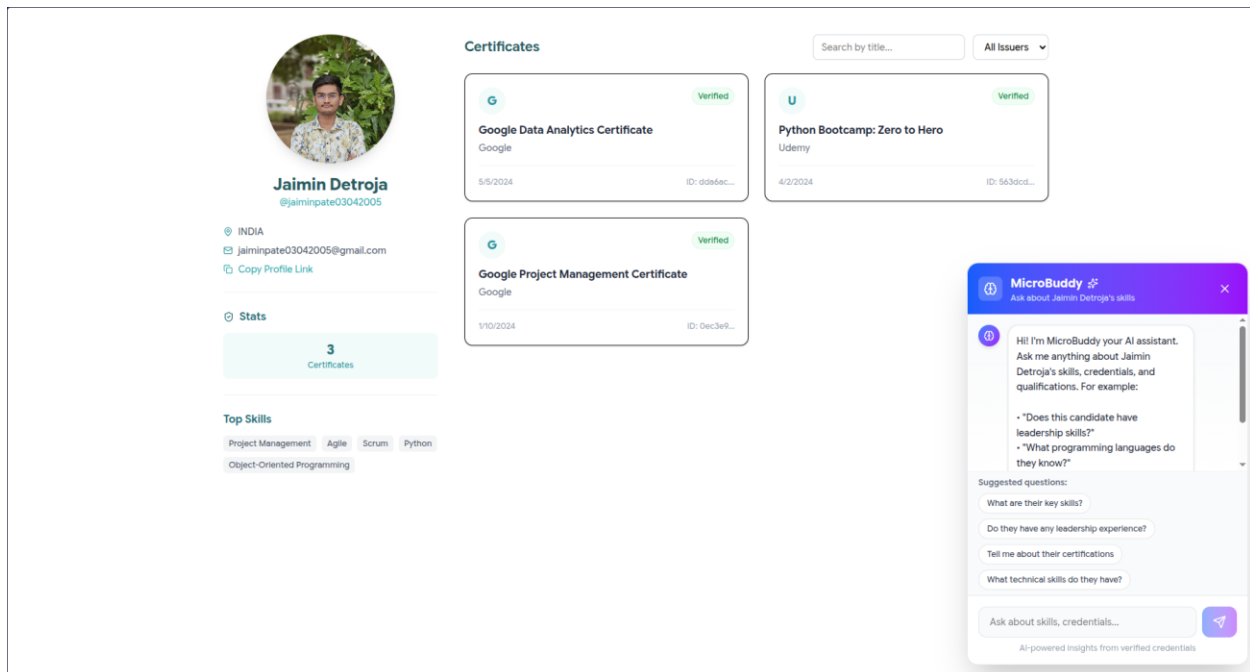


Figure 4.16: Employer AI Chatbot — Natural Language Credential Query

Single Credential Verification (/employer/verify)

The employer verification page provides the same dual-method verification as the public /verify page: a Blockchain Verify tab (accepts credential ID, tx_hash, IPFS CID, or PDF upload) and an AI Verification tab (accepts a credential ID and a scanned/photo copy of the document). Results show a VALID / INVALID badge with supporting evidence.

The screenshot shows the 'EmployerPortal' interface with a navigation bar containing 'Dashboard', 'Verification' (active), 'Find Talent', and 'Profile'. A 'Logout' link is in the top right. The main heading is 'Credential Verification' with the subtext 'Verify authenticity via ID, Transaction Hash, or IPFS CID.' Below this are two tabs: 'Single Verification' (selected) and 'Bulk Verification'. The 'Single Verification' section features a green banner stating '100% accurate and reliable' and 'Blockchain-backed cryptographic verification'. It includes a 'Credential Identifier' search bar with a 'Verify' button. Below the search bar, it provides examples of valid identifiers: 'Start typing an ID (e.g., CRED-...), a transaction hash (0x...), or an IPFS CID (Qm...)'. An alternative option is 'Or verify using file', which includes two buttons: 'Upload Credential PDF' (noting 'PDF only — credential with embedded metadata') and 'Scan with Camera' (noting 'Take a photo directly'). A section for 'AI-Powered Verification' is also present, marked as 'More flexible'. It explains that users can upload a photo or scanned copy of a marksheet or certificate and enter the Credential ID, which will be compared by Google Gemini AI against the original document stored on IPFS. At the bottom, there is a 'Credential ID' search bar and an 'Upload Document (Image or PDF)' button.

Figure 4.17: Employer Single Credential Verification

Bulk ZIP Verification

The bulk verification section allows the employer to upload a ZIP archive containing multiple credential PDFs. The system processes each PDF individually — extracting the embedded credential ID from PDF metadata and verifying checksum and blockchain record — and produces a per-file VALID / INVALID result table with download support.

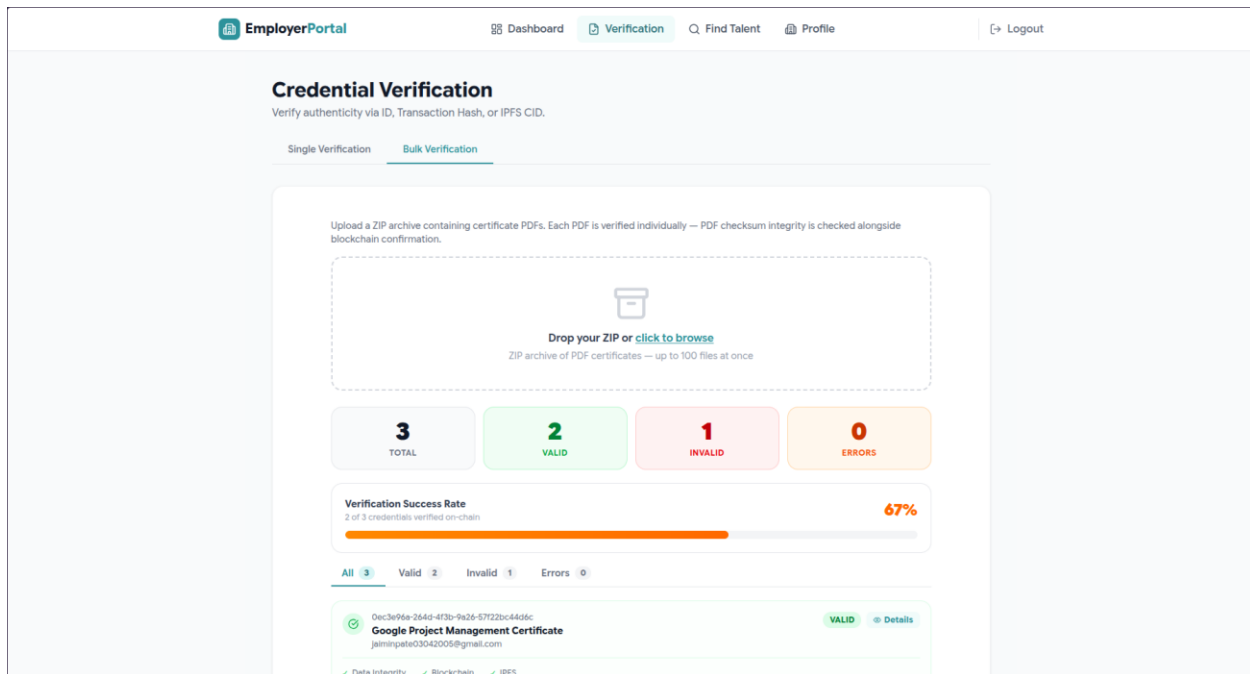


Figure 4.18: Employer Bulk Credential Verification

4.4.4 Dual Verification Interface

The public verification page is accessible to any user and implements two independent verification methods on separate tabs.

Blockchain Verification Tab

The Blockchain Verify tab accepts a Credential ID, blockchain transaction hash, or IPFS CID in a single text field, or allows the verifier to upload the original credential PDF directly. For identifier-based verification, the system rebuilds the canonical JSON hash and queries the smart contract on Sepolia, returning a result panel with `hash_match` status, `blockchain_verified` status, and an overall VALID or INVALID verdict with supporting detail. For PDF upload, the system extracts the embedded credential ID, recomputes the SHA-256 checksum, and cross-references the blockchain record.

Back to Home

Verify a Credential

Instantly verify the authenticity of certificates issued on the MicroMerit blockchain network.

Blockchain Verify
AI Verification

Verification Method
Credential Identifier

Credential ID

Verify Authenticity

Or verify using PDF

100% accurate and reliable
Blockchain-backed cryptographic verification

Upload Credential PDF

PDF only — credential with embedded metadata

Valid Credential

Verified on Sepolia Network

PDF Integrity
Blockchain Record
IPFS Storage

Credential Details

Credential ID:

563dcd07-a499-472f-bb14-c1c670ff50f5

Certificate Title:

Python Bootcamp: Zero to Hero

Issued To:

Jaimin Detroja
jaiminpate03042005@gmail.com

Issued By:

Udemy

Issued On:

April 2, 2024

IPFS CID:

QmZfqQ8C62BWBWNrJwTz8CMGH4kQqSjUPmQA6LRCzMV3uK

Data Hash:

87ace94288fcdce06f66b12eb1eb6e3b00335d305515e7f5f1eff952b2474978

Transaction Hash:

0x468e0ca80495738011a7216c556a093b948139f5093591ef01fa6359b838bbs2

View on Blockchain

View Original Certificate

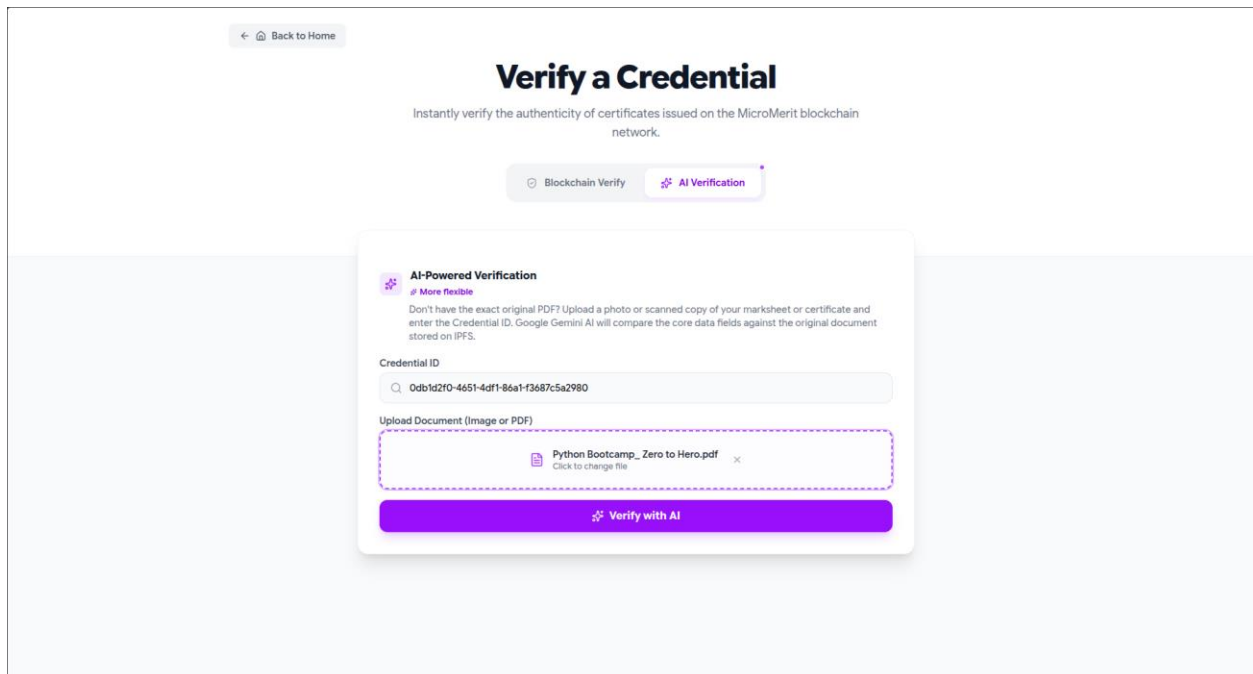
View Credential Details

Figure 4.19: Blockchain Credential Verification Interface

AI Verification Tab

The AI Verification tab accepts a Credential ID and an uploaded copy of the document (scan, photo, or printed version). The system fetches the original PDF from IPFS and sends both documents to Google Gemini 2.5 Flash for field-by-field semantic comparison. The result panel displays an overall match verdict, a confidence percentage bar, a list of matched and mismatched fields, and a one-sentence AI-generated summary. A disclaimer note clarifies that AI verification is for practical identification and that blockchain verification provides legal-grade cryptographic proof.

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The interface is titled "Verify a Credential" and includes a subtitle: "Instantly verify the authenticity of certificates issued on the MicroMerit blockchain network." At the top left, there is a "Back to Home" link. Below the title, there are two tabs: "Blockchain Verify" and "AI Verification", with the latter being the active tab. The main content area is titled "AI-Powered Verification" and includes a sub-header "More flexible". The text explains: "Don't have the exact original PDF? Upload a photo or scanned copy of your marksheet or certificate and enter the Credential ID. Google Gemini AI will compare the core data fields against the original document stored on IPFS." There is a text input field for the "Credential ID" containing the value "Odb1d2f0-4651-4df1-86a1-f3687c5a2980". Below this is a section for "Upload Document (Image or PDF)" with a dashed border and a file upload button. A file named "Python Bootcamp_ Zero to Hero.pdf" is shown with a "Click to change file" link and a close button. At the bottom, there is a large purple button labeled "Verify with AI".

Figure 4.20: AI Document Comparison Verification Interface

4.4.5 Admin Portal (Port 5174)

The Admin portal is a separate React application providing platform-wide oversight across four dashboards.

Admin Overview Dashboard

The main dashboard displays platform-wide KPI cards: total registered users (by role), total credentials issued, pending issuer applications, and credentials awaiting blockchain confirmation. A recent-activity feed shows the latest issuances and registrations across the platform.

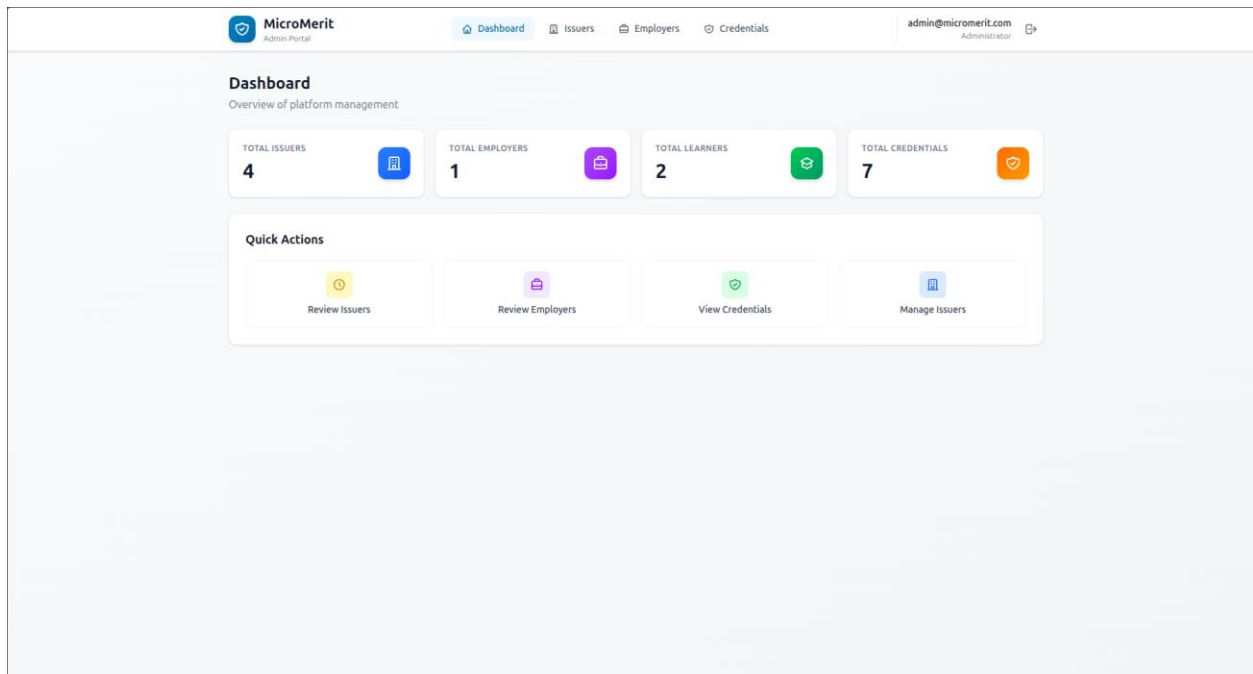


Figure 4.21: Admin Portal — Overview Dashboard

Issuer Management Dashboard

The Issuers dashboard lists all registered issuer accounts with their approval status (Pending / Approved / Rejected). Each row has action buttons to approve or reject the application with an optional reason. Rejected issuers are blocked from accessing the issuance form until re-approved.

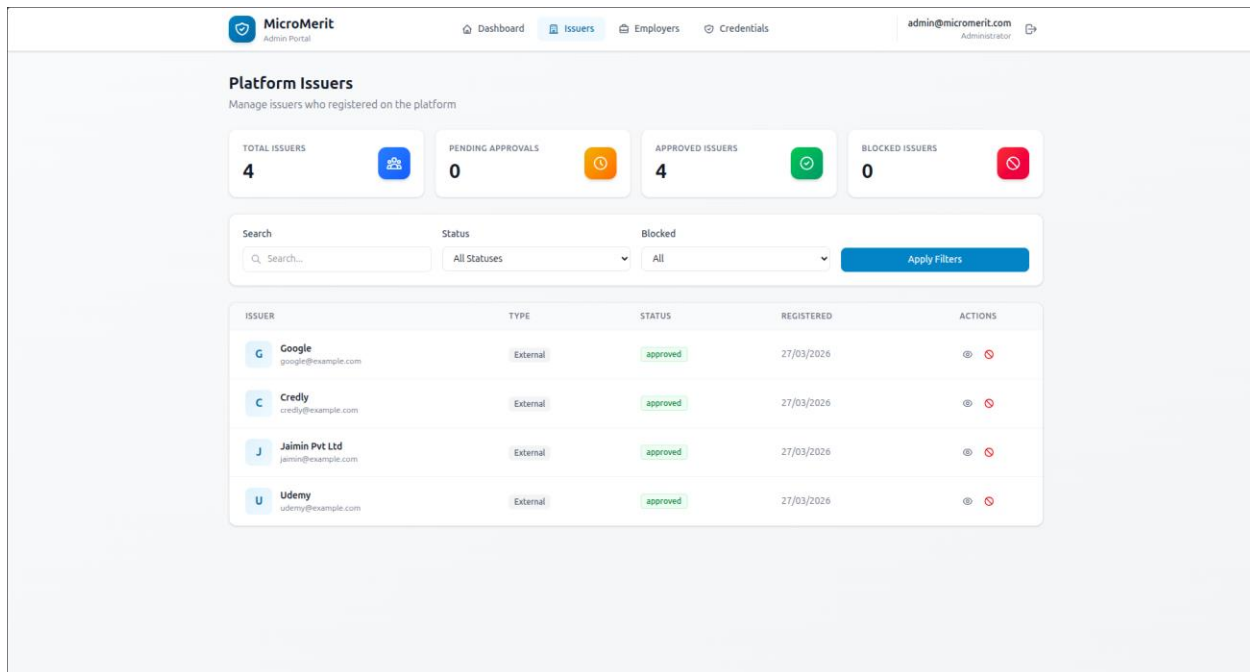


Figure 4.22: Admin Issuer Approval and Management

Credential Oversight Dashboard

The Credentials dashboard provides an admin-level view of all credentials across all issuers, with filters for blockchain status, issuer, and date range. Admins can inspect individual credential records, view blockchain and IPFS details, and flag credentials if needed.

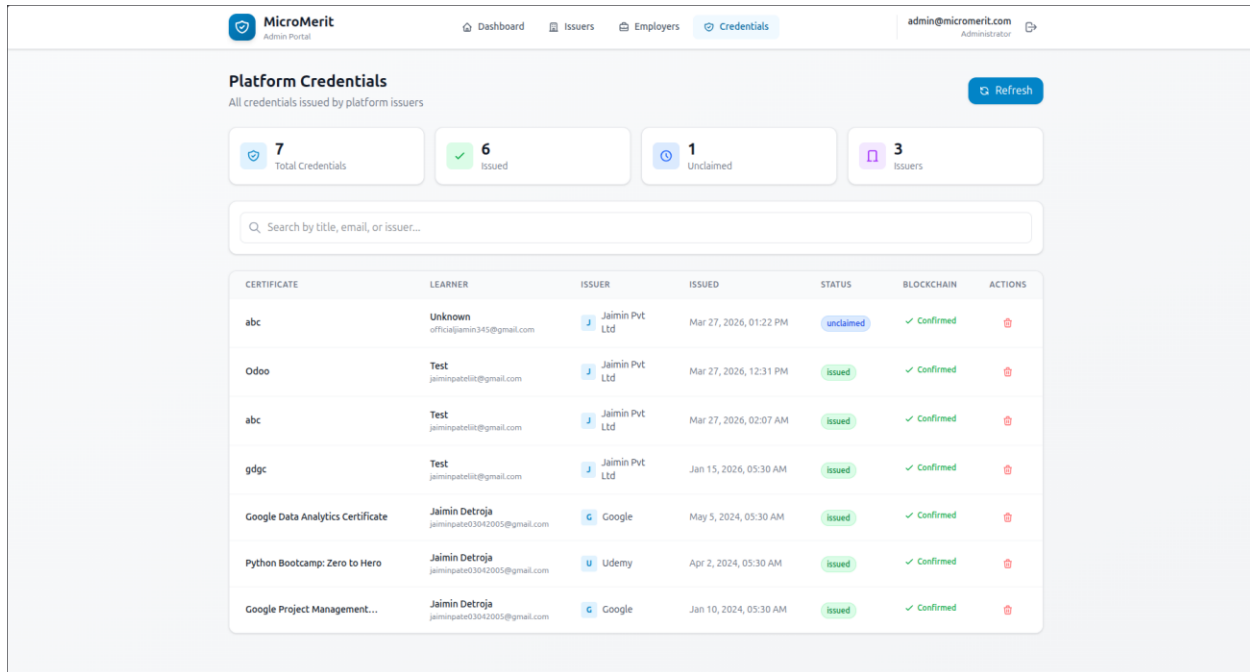


Figure 4.23: Admin Credential Oversight Dashboard

4.5 Results and Output

The system was successfully implemented and tested across all functional requirements. Single and bulk credential issuance operate correctly, with blockchain anchoring completing asynchronously and status updating from Pending to Confirmed after on-chain confirmation. The dual-method verification correctly detects tampered PDFs: modifying even a single byte of a credential PDF causes the checksum comparison to fail. Blockchain verification correctly returns VALID for genuine credentials and INVALID for tampered or fabricated ones.

The AI verification module successfully identifies genuine credentials from scanned copies and photographs, returning high confidence scores (90–95%) for clean scans. The employer AI chatbot correctly responds to natural-language queries about learner skills and qualifications. The external sync module successfully fetches and normalizes credentials from the mock provider, with the pending-claim mechanism correctly resolving upon learner registration. The admin portal correctly gates issuer registration — issuers cannot issue credentials until their application is approved.

CHAPTER 5: CONCLUSIONS AND FUTURE SCOPE

5.1 Conclusion

MicroMerit Portal successfully demonstrates the viability of combining blockchain immutability, IPFS decentralized storage, and AI intelligence into a unified credential management platform. The project addresses five core problems — credential forgery, manual verification overhead, lack of aggregation across providers, absence of employer-facing AI tools, and inaccessible learner profiles — with concrete, working technical solutions.

The blockchain anchoring mechanism provides a cryptographically verifiable chain of custody for every issued credential. The canonical JSON approach — where all keys are sorted alphabetically before hashing — ensures that the `data_hash` is deterministic and reproducible by any party, enabling trustless verification without requiring access to the original system. The IPFS storage layer ensures that credential PDFs remain permanently accessible at a content-addressed URL, independent of the platform's availability.

The AI components add a layer of practical flexibility that pure cryptographic systems lack. The Tesseract + Groq OCR pipeline automates credential metadata extraction, reducing issuer effort. The Google Gemini document comparison enables verification of physical certificate copies — a common real-world requirement — that would fail under byte-level hashing. The employer AI chatbot demonstrates how structured credential data can be used as context for natural-language querying.

From an architectural perspective, the microservices design proved effective in isolating concerns. The connector pattern for external sync made it straightforward to add new credential providers without modifying the core issuance logic. The adapter pattern for bulk issuance provides a clean extension point for supporting additional file formats in the future. Overall, the project fulfilled its stated objectives and is a functional, demonstrable system.

5.2 Limitations

The current implementation has the following limitations which are acknowledged:

- The blockchain integration uses the Sepolia testnet; production deployment would require real ETH for gas fees and a mainnet contract deployment.
- AI verification is probabilistic and cannot be used as legal proof of credential authenticity.
- The external credential sync connectors use a mock provider server for development; integration with live Udemy, Credly, or other provider APIs would require production API access and legal agreements.
- The bulk issuance adapter currently supports only JSON files; PDF and image OCR adapters are scaffolded but not fully implemented.

5.3 Future Scope

1. **Skill Pathway AI (Roadmap Generation):** Based on a learner's current credential portfolio, an AI agent would analyze skill gaps and recommend the next credentials to obtain, generating a personalized learning roadmap. The backend scaffolding and database schema (LearnerRoadmap table) for this feature already exists in the codebase.
2. **AI-Generated Skill Profiles:** The LearnerSkillProfile table already exists in the schema. This feature would automatically generate a structured skill profile from all verified credentials, summarizing the learner's competencies and domain expertise in a machine-readable format for use by employer ATS (Applicant Tracking Systems).
3. **Extended Provider Connectors:** Real API integrations with Udemy, Coursera, and other credential platforms would be implemented, replacing the mock provider server. The connector pattern design makes this a straightforward extension.

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Note: If your in-text citations currently reference [1]–[18], keep the numbering consistent. If you want, the in-text citations can be updated to match this revised list.